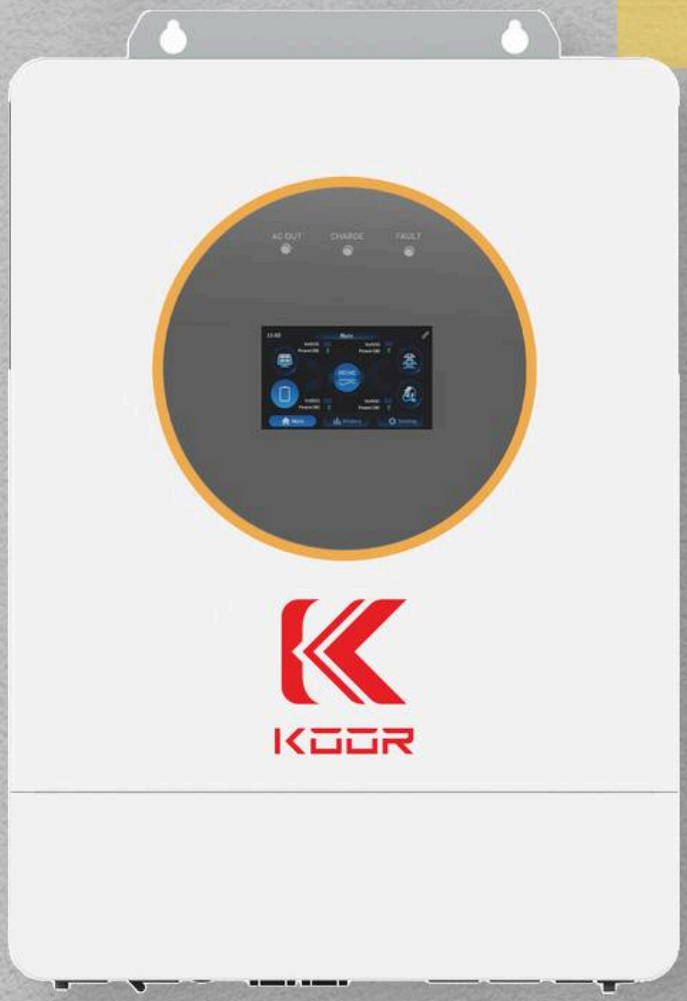
Off-Grid Inverter

KR-A12US

12 kW Split Phase



KR-A12US

12 kW OFF-GRID

120/240V SPLIT PHASE

DUAL MPPT | 500V PV

User Manual

READ THE INSTRUCTIONS COMPLETELY BEFORE OPERATING THE EQUIPMENT

Check the utility voltage before turning ON the unit.

Verify the inverter's programmed grid type before connecting to the utility.

The unit is programmed for 120/240V split phase at 60Hz by default.

Disregarding these instructions could result in permanent damage to the unit.



DISCLAIMER

UNLESS SPECIFICALLY AGREED TO IN WRITING:

KOOR assumes no responsibility or liability for any damages, property loss, personal injury, or any adverse consequences resulting from improper use and installation of the product or the failure to adhere to the guidelines provided in this document. Users are expressly advised to follow the instructions and guidelines outlined in the documentation accompanying the product. KOOR shall not be liable for any damages or losses incurred due to deviations from recommended usage, installation, or maintenance procedures. By using the product, users acknowledge their understanding of these disclaimers and agree to use the product at their own risk. KOOR reserves the right to update or modify product information, specifications, and guidelines without prior notice.

KOOR retains the right to final interpretation of this document and all related materials pertaining to this product. This document is subject to modifications, updates, revisions, or termination without prior notice.

Fantastech Energy Solution LLC

Email: derek@fantastech.org | Phone: 574-218-1973 | Goshen, IN 46528, US

12 kW Split Phase Off-Grid Inverter — Quick Start

KR-A12US | 120/240 V split phase, 60 Hz | off-grid only

Keep this sheet with the inverter. It covers the steps you use on the job. Full detail, every setting and the whole fault list are in the User Manual in this same book. Qualified electrical personnel only.

1 WHAT IS ON THE UNIT

PORTS AND PARTS

1	AC output port
2	AC input port
3	WiFi port (optional)
4	USB communication port
5	ON/OFF rocker switch
6	Photovoltaic (PV) terminals
7	Battery terminals
8	CT current sensing for self-consumption (optional)
9	Dry contact port
10	Parallel communication port (parallel model only)
11	RS485-2/BMS communication port
12	WiFi port
13	AC input breaker
14	Grounding screw hole

SCREEN AND CONTROLS

Screen	The screen is a resistive touch LCD. There are no press buttons. You tap the screen.
Main	Main page tab. Shows live data. Press the screen to scroll through the machine's real-time data.
History	Historical data tab.
Setting	Press the "Setting" button to open the setup menu. Menu groups: Normal, Battery, Parallel, BMS, Hybrid and Others.
< and >	Press "<" and ">" to select the value you want.
Confirm	Press "Confirm" to save the new setting.
ON/OFF rocker switch	On the bottom of the case. Press it to turn the unit on. The screen and the indicator light up.

2 WHERE IT CAN HANG

Max ambient temp	55°C (131°F)
Min ambient temp	-10°C (14°F)
Operating temperature range	-10°C to 55°C (14°F to 131°F) (derating above 45°C / 113°F)
Storage temperature range	-25°C ~ 60°C (-13°F ~ 140°F)
Humidity range	5% to 95% (conformal coating protection)
Clearance above and below	50 cm (19.7 in)
Clearance each side	20 cm (7.9 in)
Size (L*W*D)	620mm*450mm*132mm (24.41 in × 17.72 in × 5.20 in)
Weight	23.8 kg (52.47 lb)
Noise	≤60dB
Cooling	Forced air cooling, variable fan speed

YES

- Mount the unit vertically on the wall. This is the recommended installation position.
- Mount it on a sturdy surface.
- SUITABLE FOR MOUNTING ON CONCRETE OR OTHER NON-COMBUSTIBLE SURFACE ONLY.
- Install the inverter at eye level so the LCD display can be read at all times.
- Leave 50 cm (19.7 in) above and below, and 20 cm (7.9 in) on each side. This gives enough room for heat to escape and enough space to remove wires.
- Install the unit with four screws. M4 or M5 screws are recommended.
- We recommend that you install a proper fuse or circuit breaker outside the inverter.

NO

- Do not install the inverter on flammable building materials.
- Do not install the inverter in harsh places, such as areas that are damp, oily, flammable, explosive, or very dusty.
- Do not take the unit apart. When service or repair is needed, take it to a qualified service center.
- Do not connect one PV string to more than one inverter at the same time.

3 MOUNTING

1. Inspect the unit. Make sure nothing inside the package is damaged.
2. Take off the bottom cover before you connect any wiring. Remove the four screws.
3. Pick the spot: vertical on a sturdy, non-combustible wall, at eye level.
4. Hold the clearances: 50 cm (19.7 in) above and below, 20 cm (7.9 in) each side.
5. Mark the holes. Mounting hole layout and dimensions: 230 mm, 92 mm and 555.5 mm; holes Ø18 mm, Ø8 mm and Ø6 mm.
6. Install the unit with four screws. M4 or M5 screws are recommended.
7. After all wiring is connected, put the bottom (base) cover and the dust covers back on.

4 WHAT IS IN THE BOX

The unit	1
User manual	1
Ring terminal	2
Expansion bolts	2
Expansion screws	2
Dust cap	2
Parallel communication line (listed in section 8.1, not in the section 3.1 packing list)	1

Check the box and the unit for damage before you sign for it. Note any dents, holes or cracks on the freight paperwork.

5 CABLES

Battery	67mm²/00AWG	Recommended circuit breaker 2P—300A
AC input	13mm²/6AWG	Recommended circuit breaker 3P—63A. Maximum bypass input current 60A.
AC output	13mm²/6AWG	Recommended circuit breaker 3P—63A. Maximum bypass output current 60A.
PV	5.2mm²/10AWG	Recommended circuit breaker 2P—32A. Maximum PV input current 27A.
AC strip length	Strip 10 mm (0.39 in) of insulation from each of the eight conductors	Shorten the phase conductor L and the neutral conductor N by 3 mm (0.12 in)
PV strip length	Strip 10 mm (0.39 in) from the + and − conductors	
Battery BMS	RJ45 cable	Battery RS485/CAN jack to the inverter RS485-2/BMS port
Parallel communication	Standard DB15 computer cable, shielded	Male connector (out) of one inverter to the female connector (in) of the next

TORQUE

Battery terminal bolts	2-3Nm
AC input terminals	1.2-1.6Nm
AC output terminals	1.2-1.6Nm
PV terminals	1.2-1.6Nm

BEFORE YOU TOUCH A TERMINAL

- Read all instructions and warning marks on the unit, on the batteries, and in this manual before you use the unit.
- All operation and connection work must be done by a professional electrical or mechanical engineer.
- To reduce the risk of electric shock, disconnect all wiring before you do any maintenance or cleaning. Turning off the unit will not reduce this risk.
- Make sure all power sources to the inverter are disconnected and all capacitors are discharged before you check or work on the unit.
- Install a separate DC over-current protector or disconnect device between the battery and the inverter.
- Install a separate AC breaker between the inverter and the AC input power source. Make sure the AC power source is disconnected before you hardwire it to the unit.
- Install a separate DC circuit breaker between the inverter and the PV modules.
- Open (turn off) the DC protector or disconnector first, before you make the AC input/output connection.
- Before you make the final DC connection or close the DC breaker, make sure positive (+) is connected to positive (+) and negative (−) is connected to negative (−). Do not reverse the polarity of the battery input.
- NEVER short-circuit the AC output or the DC input.
- Shock hazard. Batteries connected in series have a high voltage.
- There are two terminal blocks, marked "IN" and "OUT". Do NOT mix up the input and output connectors.
- Be sure to connect the PE (protective earth, ground) conductor first.
- Decide first which battery system you want: lithium or lead-acid. The wrong setting keeps the system from working normally.

6 PV CHECK BEFORE YOU LAND IT

Max PV array open circuit voltage	500Vdc
Start-up voltage	120Vdc
MPPT voltage range	90Vdc~450Vdc
PV operating voltage range	90Vdc~485Vdc
Max PV input current	27A + 27A
Max PV input power	9000W + 9000W
Number of MPPT groups	2
Recommended PV wire size	5.2mm²/10AWG
PV charge current (can be set)	250A
PV module rule 1	The open circuit voltage (Voc) of the PV array must not be higher than the inverter's maximum PV array open-circuit voltage.
PV module rule 2	The open circuit voltage (Voc) of the PV array should be higher than the start-up voltage.

Meter every string open circuit before you land it, and correct for the coldest day of the year. Voc must stay under the maximum PV array open circuit voltage above.

7 COMMUNICATION PORTS

USB-B port	USB communication with the optional PC host software. Install the "USB to serial chip CH340T driver" on the computer.
WiFi port	WiFi module, optional. Two types to choose from: a USB Type-A port or an M14 aviation waterproof connector port with antenna.
RS485-2/BMS port	Talks directly with the lithium battery BMS over RS485 (custom), or over CAN on pins 4 and 5.
Dry contact port	Terminals IP, GND, N, NO, NC. N-NO-NC can turn a diesel generator on and off. IP-GND turns the inverter AC output on and off by remote control: open = AC output ON, closed = AC output OFF.
CT sampling detection (anti feed-in)	Optional. Terminals CT-L2 (− +) and CT-L1 (− +). Measures the input line current for the self-use function. Use only the CT sensors supplied for this inverter. The CT arrows point to the inverter.
Parallel communication port	Parallel communication between units. Each inverter has two DB15 ports, one male and one female. Never connect an inverter's male connector to its own female connector.

BATTERY BMS PORT PINOUT

1	RS485-B
2	RS485-A
3	not used
4	CAN-H
5	CAN-L
6	not used
7	RS485-A
8	RS485-B

8 TURN IT ON — IN THIS ORDER

1. Close (turn on) the battery circuit breaker.
2. Press the rocker switch at the bottom of the inverter. The screen and the indicator light up. This shows the inverter is now on.
3. Close the circuit breakers for PV, AC input and AC output, in that order.
4. Turn on the loads one at a time. Start with the smallest load and work up to the largest.

9 SHUT IT DOWN

1. The On/Off switch is on the bottom of the case (section 4.1, Power ON/OFF).
2. To turn off only the AC output from the screen: Setting > Others > "On/OFF control" > "Shut down". "Start up" turns the inverter AC output back on.
3. To restart the inverter from the screen: Setting > Others > "Restart control" > "Restart".
4. Turning off the unit does not make it safe to work on. Disconnect all wiring before any maintenance or cleaning.
5. Before any maintenance work: make sure all power sources to the inverter are disconnected and all capacitors are discharged.
6. To add an inverter to a running parallel system: disconnect the battery input, PV input, AC input and AC output, and make sure every inverter is powered off first.
7. Turn the inverter OFF and switch the battery off before you plug in or unplug a communication cable.

10 KOOR BATTERY COMMUNICATION

1. Turn the inverter OFF and switch the battery off before you plug in or unplug a communication cable.
2. Set the master battery's four address switches all OFF (address 0). Up is ON. Set the switches with the battery off.
3. Give every other battery its own address: 1, 2, 3 and up to 14. No two batteries share an address. Changed a switch with the battery on? Hold RST 3 s, then 3 s again.
4. Only one battery talks to the inverter. Run a plain straight ethernet cable (CAT5e or CAT6) from its CAN jack to the inverter's RS485-2/BMS port.
5. Active Stack and 16K Pro wire the same. The CAN jack is the right-hand jack of the RS485-1/CAN pair. Pin 4 is CAN-H and pin 5 is CAN-L on both ends. No crossed or custom cable.
6. Chain the rest in a line on RS485-2: out of B on one battery, into A on the next. The master's A jack and the last battery's B jack stay empty.
7. Never run a cable from the last battery back to the master.
8. Set Pylon on the battery too: every pack's CAN protocol = Pylon (Active Stack: battery app, password 1234), then RST 3 s, then 3 s again.
9. On the inverter set "Battery type" to LFP16 (16 cells in series, 51.2V nominal).
10. Set "RS485-2" to CAN-BMS.
11. Set "BMS protocol" to Pylon Can. Then save.
12. If there is still no battery data: move the cable to the battery's RS485-1 jack (backup path, plain straight cable too), then set "RS485-2" to 485-BMS and "BMS protocol" to Pylon.

Full battery detail — port, address switches, settings and what to do when it will not talk — is in Appendix A of the User Manual in this book.

11 POST-INSTALL CHECK

- ☐The screen shows the battery state of charge as a percent from the BMS, with battery voltage and current.
- ☐No BMS fault code is showing.
- ☐Charge current follows the battery's own limit instead of the value you set by hand.
- ☐Measure the output voltage and confirm it is correct. Then connect the loads.
- ☐Make sure the wires are securely connected, and that all wiring is fastened so no wire comes loose in use.
- ☐Check that the indicators and display match how the unit is working.
- ☐Check all wiring terminals for corrosion, damaged insulation, signs of high heat, or burning/discoloration. Tighten the screws.
- ☐Make sure nothing blocks the airflow around the unit.

12 TOP FAULTS

Code	Fault	First thing to check
/	No display on the screen	Check that the battery circuit breaker or the PV circuit breaker is closed. Check that the switch is in the "ON" position. Tap the screen to wake it.
[05]	Battery not connected	Check whether the battery is not connected, or whether the battery circuit breaker is not closed.
[01] [02]	Battery under voltage protection	Charge the battery until it reaches the low voltage disconnection recovery voltage.
[06]	Battery over voltage	Measure the battery voltage to see if it is above the rated value. If it is, discharge the battery until its voltage is below the battery over-voltage recovery point.
[18]	PV over-voltage	Use a multimeter to check whether the PV input voltage is above the rated maximum allowed input voltage.
[09] [10]	Bypass overload protection, inverter overload protection	Use fewer power devices (reduce the load). Then restart the unit to restore load output.
[11]	Inverter short circuit protection	Check the load wiring carefully and clear any short-circuit points. Then power the unit up again.
[22] [23]	Heat sink over temperature protection	When the unit cools below the recovery temperature, normal charge and discharge control starts again.
[27]	Fan failure	Check whether the fan is not turning or is blocked by a foreign object.
[49]	BMS communication error	Check that the communication line is connected correctly. Check that the BMS is set to the matching lithium battery communication protocol.
[40] [42]	Parallel connection fault	Check whether the parallel cable is poorly connected, for example loose or wired wrong.
[37]	Parallel ID setting error	Check whether the same parallel ID number is set more than once.

Full fault list, settings and wiring diagrams are in the User Manual in this book. Questions before you energize it — call KOOR support at 574-218-1973.

Limited Warranty — Terms

KOOR KR-A12US 12 kW Split Phase Off-Grid Inverter · sold and warranted by Fantastech Energy Solution LLC

WHAT IS COVERED AND FOR HOW LONG

KOOR off-grid inverters — 5 years. KOOR batteries — 10 years. The term runs from the date of installation, and in no case more than six months past that term from the date the unit left the factory.

The warranty covers defects of design, components and manufacturing. It applies to units bought from Fantastech and installed at the address on the warranty card. It is non-transferable.

WHAT IS NOT COVERED

- Incorrect installation or commissioning.
- Failure to follow the User Manual, the installation guide or the maintenance instructions.
- Unauthorized modifications, changes or attempted repairs.
- Incorrect use or inappropriate operation, including reversed battery polarity and use of the wrong battery communication protocol.
- Insufficient ventilation of the device.
- Failure to observe applicable safety regulations and electrical code.
- Breaking the product seal — opening the casing — without written approval.
- Grid-tied use. These are off-grid inverters and are not certified for grid connection.
- Force majeure: lightning, over-voltage, storm, flood, fire.
- Transport damage, and superficial or cosmetic defects — dents, marks, scratches — that do not affect how the inverter works.

HOW A CLAIM WORKS

- Report it within 14 days of the first sign of a problem. Call 574-218-1973 or email derek@fantastech.org.
- Send the completed warranty card, the purchase invoice, the installation or commissioning report, and any extension warranty card if you have one.
- Send pictures and the fault code or fault screen. It is the fastest way to get a unit moving.
- Fantastech decides whether the unit is repaired on site, repaired at the shop, or exchanged for a replacement device of equivalent value for model and age.
- On an exchange, the remainder of the original warranty carries over to the replacement device. A replacement does not start a new warranty term and does not come with a new certificate.
- Because of ongoing product improvement, a replacement part or unit may not match your system monitoring or other components, and the cost of changing those out is not part of this warranty.

LIMITS

- This warranty covers the equipment. It does not cover labor, travel, lift or crane rental, lost production, lost profits, or other direct or indirect costs, unless Fantastech agrees to them in writing beforehand.
- Claims beyond the rights stated here — including costs of removal and reinstallation — are not covered.
- Your rights under statutory liability and product liability law are unaffected by this warranty.

Keep the warranty card and your invoice together. Both are needed to support a claim. If the card was never sent in, send it with the claim.

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1. Information on This Manual

This manual explains how to assemble, install, operate and troubleshoot this unit. Read this manual carefully before you install or use the unit. Keep this manual for future use.

1.1 Scope

This manual gives safety and installation guidance. It also gives information about tools and wiring.

1.2 Target Group

This manual is for qualified persons and end users. End users may also do tasks that do not need any special qualification. Qualified persons must have these skills:

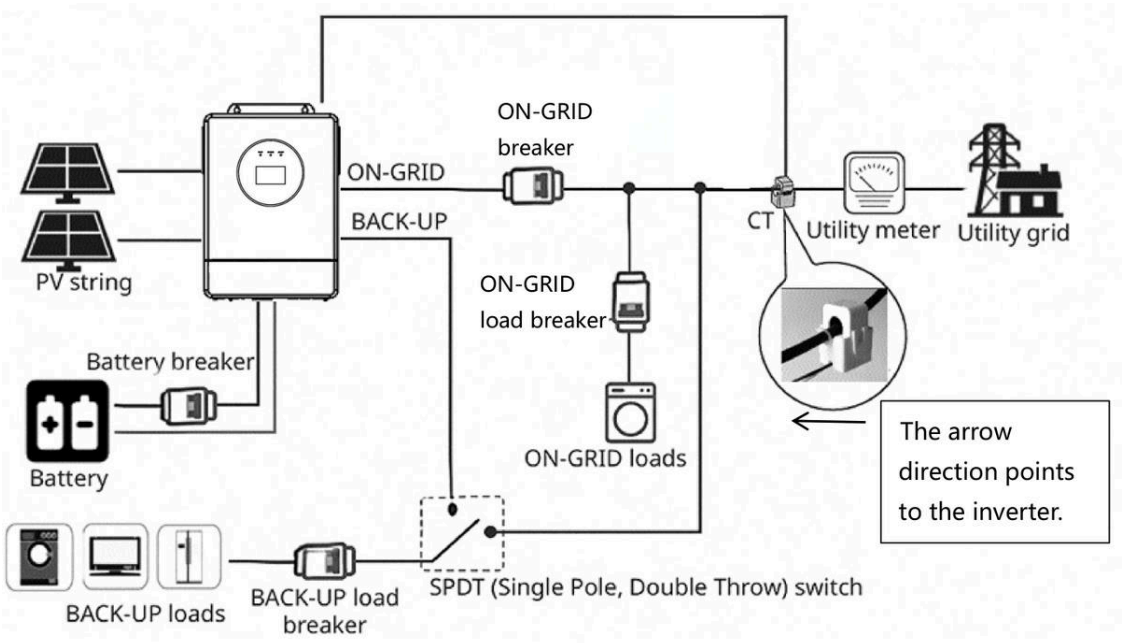
1. Knowledge of how an inverter works and how it is operated.
2. Training in how to deal with the dangers and risks of installing and using electrical devices and installations.
3. Training in how to install and commission (start up) electrical devices and installations.
4. Knowledge of the standards and directives that apply.
5. Knowledge of this manual and all safety information, and following them.

1.3 Safety Instructions

1. Before you use the unit, read all instructions and warning marks on the unit, on the batteries, and in all related sections of this manual. If the unit is not installed as this manual says and equipment damage results, KOOR has the right to refuse quality assurance (warranty) coverage.
2. Be sure which kind of battery system you want: a lithium battery system or a lead-acid battery system. If you choose the wrong system, the energy storage system cannot work normally.
3. Do not take the unit apart. When service or repair is needed, take it to a qualified service center. If the unit is put back together wrong, it can cause electric shock or fire.
4. All operation and connection work must be done by a professional electrical or mechanical engineer. All electrical installation must meet local electrical safety standards.
5. **CAUTION:** To reduce the risk of injury, charge only deep-cycle lead-acid rechargeable batteries and lithium batteries. Other types of batteries may burst and cause personal injury and damage.
6. To reduce the risk of electric shock, disconnect all wiring before you do any maintenance or cleaning. Turning off the unit will not reduce this risk.
7. For best operation of this inverter, follow the required specifications when you choose cable sizes. Correct cable size is very important for correct operation of this inverter.
8. **GROUNDING INSTRUCTIONS:** This inverter should be connected to a permanent grounded wiring system. Follow local requirements and regulations when you install this inverter.
9. **NEVER** short-circuit the AC output or the DC input. Do NOT connect to the grid/utility (mains) when the DC input is short-circuited.
10. Do not install the inverter in harsh places, such as areas that are damp, oily, flammable, explosive, or very dusty.

11. **Warning!** Only qualified maintenance personnel may repair this equipment. If the fault is still there after you follow the troubleshooting table, take this inverter/charger back to your local dealer or service center for repair.
12. Do not reverse the polarity of the battery input. Reversed polarity may damage the unit or cause risks that cannot be predicted.
13. Make sure the inverter is fully assembled before you operate it.
14. Make sure the inverter's AC output is the only power source feeding the loads. Do not run it in parallel with any other AC power source. This can cause damage.
15. We recommend that you install a proper fuse or circuit breaker outside the inverter.

1.4 Introduction



System application example. The arrow on the CT points to the inverter.

The figure above shows how this product is used in a system. The exact system wiring depends on your actual site. A complete system has these parts:

1. **PV string:** Turns light into DC power. This power charges the battery through the inverter, or the inverter turns it directly into AC power to run the loads.
2. **Grid/utility (mains) or generator:** Connects to the AC input. It powers the loads and charges the battery at the same time. If no grid or generator is connected, the system can still run normally. The loads are then powered by the battery and the PV modules.
3. **Battery:** Stores and releases energy as the system needs. The unit supports lithium batteries and lead-acid batteries (chosen in the settings).
4. **Household loads:** You can connect many kinds of home and office loads, including refrigerators, lamps, TVs, fans and air conditioners.
5. **Inverter:** The energy conversion unit of the whole system.
6. **WiFi or GPRS:** Allows remote monitoring of the system.
7. **CT (current transformer):** Measures current. Two CTs are placed on the two hot (L1/L2) conductors. The CT arrows point to the inverter.

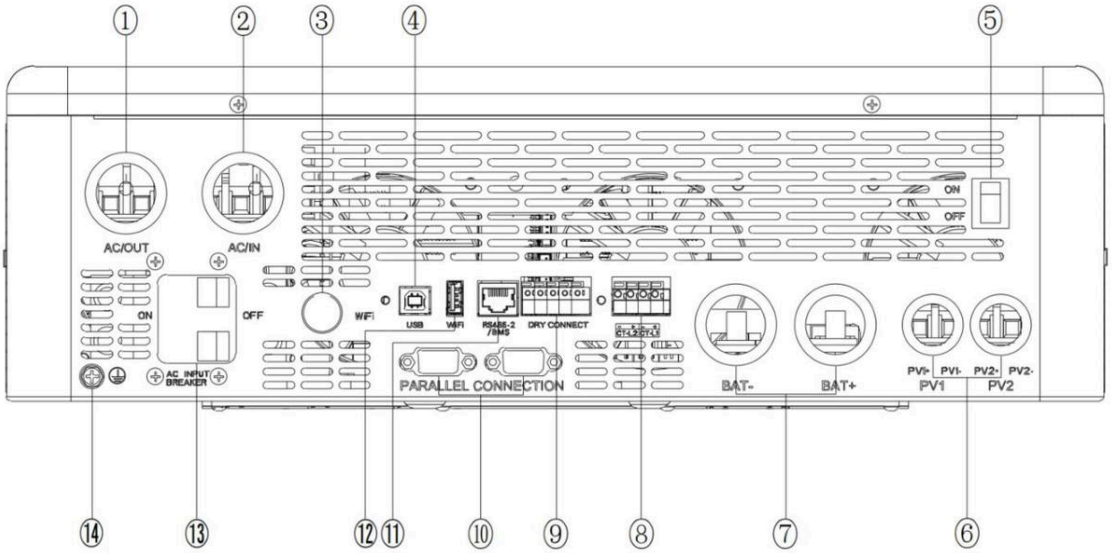
1.5 Product Features

1. Time-period (sectional) charge and discharge: you can set the time periods when the battery charges or discharges.
2. Works with lead-acid batteries and lithium batteries.
3. Anti feed-in and self-consumption function (loads can be powered by a mix of solar, battery and grid).
4. Can be used without a battery. Solar and the utility grid can power the loads at the same time.
5. Dual activation of a lithium battery: when the lithium-ion battery is dormant (asleep), either grid/utility (mains) power or solar power can wake it up.
6. Power saving mode is available to reduce no-load loss.
7. Four charging modes: Only Solar, Mains Priority, Solar Priority, and Mains & Solar hybrid charging.
8. Two output modes: mains bypass and inverter output. Uninterrupted power supply.
9. Complete protection, including over-voltage and under-voltage protection, overload protection, reverse protection, and more. 360° all-round protection with many protection functions.
10. Resistive touch LCD screen with a moving power-flow diagram, so system data and operating status are easy to understand.
11. Fully digital voltage and current double closed-loop control and advanced SPWM technology give a pure sine wave output. Supports split-phase and single-phase pure sine wave output. Supports five voltage levels per phase: 100Vac, 105Vac, 110Vac, 120Vac and 127Vac.
12. Two independent MPPTs. Advanced MPPT technology with a tracking efficiency of 99.9%.
13. Smart variable-speed fan cools the unit well and extends system life.
14. Many ways to interact with the unit: supports CAN, USB, WiFi and RS485 communication, local and remote power on/off, and dry contact control.
15. Expandable: an external CT allows whole-house anti feed-in and self-consumption.

2. Product Overview



LCD touch screen, Main page (tabs: Main, History, Setting)



Bottom connection panel (numbers match the table below)

No.	Part	No.	Part
①	AC output port	⑧	CT current sensing for self-consumption (optional)
②	AC input port	⑨	Dry contact port
③	WiFi port (optional)	⑩	Parallel communication port (parallel model only)
④	USB communication port	⑪	RS485-2/BMS communication port
⑤	ON/OFF rocker switch	⑫	WiFi port
⑥	Photovoltaic (PV) terminals	⑬	AC input breaker
⑦	Battery terminals	⑭	Grounding screw hole

3. Installation

3.1 Unpacking and Inspection

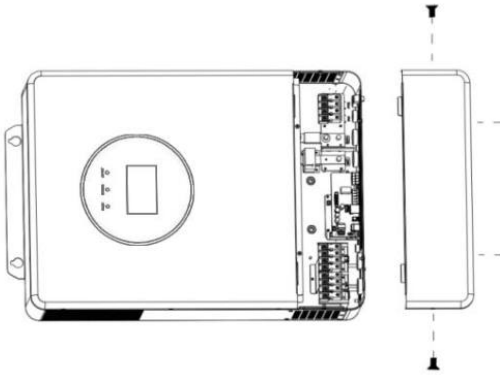
Before you install the unit, inspect it. Make sure nothing inside the package is damaged.

The package should contain these items:

- The unit x 1
- User manual x 1
- Ring terminal x 2
- Expansion bolts x 2
- Expansion screws x 2
- Dust cap x 2

3.2 Preparation

Before you connect any wiring, take off the bottom cover. Remove the four screws as shown below.

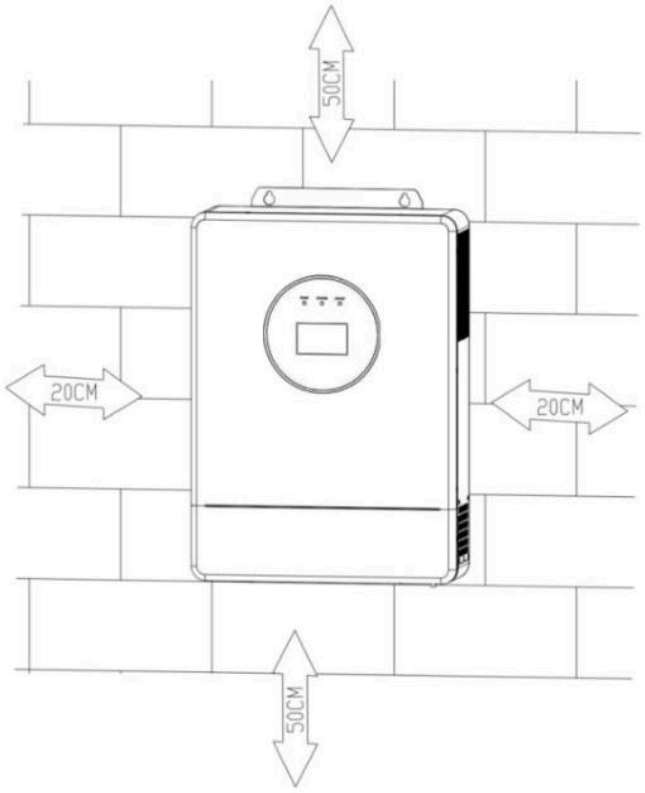


Remove the four screws to take off the bottom cover.

3.3 Mounting the Unit

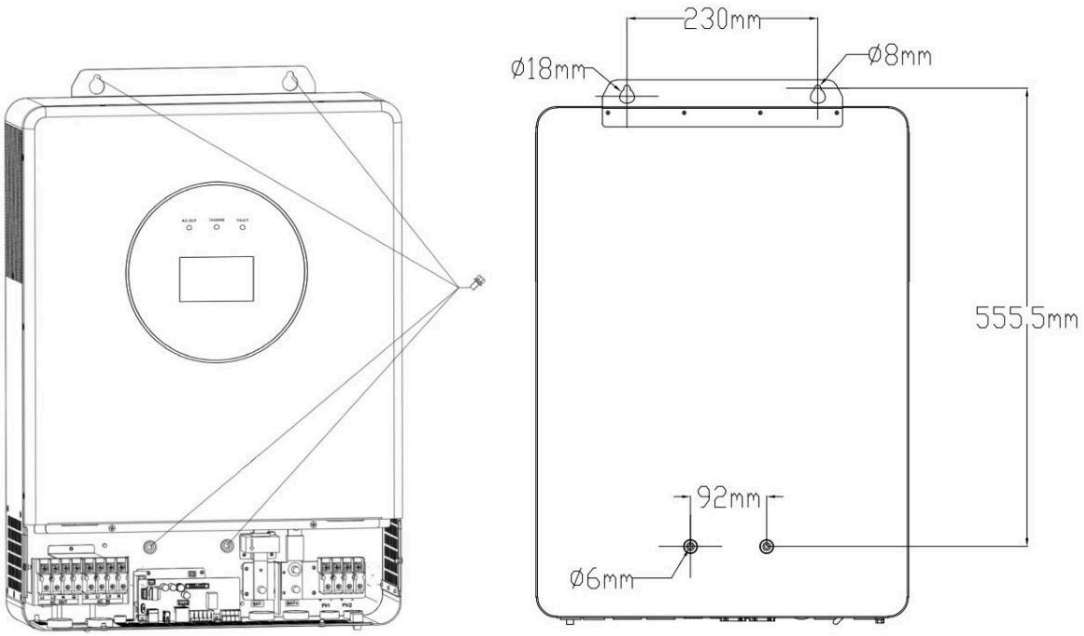
Think about these points before you choose where to install the unit:

- Do not install the inverter on flammable building materials.
- Mount it on a sturdy surface.
- Install the inverter at eye level so the LCD display can be read at all times.
- For best operation, the ambient (surrounding air) temperature should be between -10°C and 55°C (14°F and 131°F).
- Mount the unit vertically on the wall. This is the recommended installation position.
- Keep other objects and surfaces at the distances shown in the diagram. This gives enough room for heat to escape and enough space to remove wires.



Clearance around the unit: 50 cm (19.7 in) above and below, 20 cm (7.9 in) on each side.

WARNING **SUITABLE FOR MOUNTING ON CONCRETE OR OTHER NON-COMBUSTIBLE SURFACE ONLY.**



Mounting hole layout and dimensions: 230 mm, 92 mm and 555.5 mm; holes Ø18 mm, Ø8 mm and Ø6 mm.

Install the unit with four screws. M4 or M5 screws are recommended.

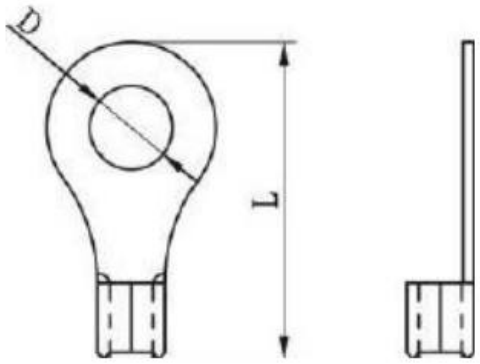
3.4 Lead-acid Battery Connection

You can use a lead-acid battery of the right capacity with a nominal voltage of 48V. You also need to set the battery type to "GEL (default)", "SLD" or "FLD".

CAUTION For safe operation and to meet regulations, install a separate DC over-current protector or disconnect device between the battery and the inverter. Some installations may not require a disconnect device, but over-current protection is still required. Use the typical amperage in the table below to choose the fuse or breaker size.

WARNING All wiring must be done by a qualified person.

WARNING Using the right cable for the battery connection is very important for system safety and efficient operation. To reduce the risk of injury, use the recommended cable and terminal size shown below.



Ring terminal (D = hole diameter, L = length).

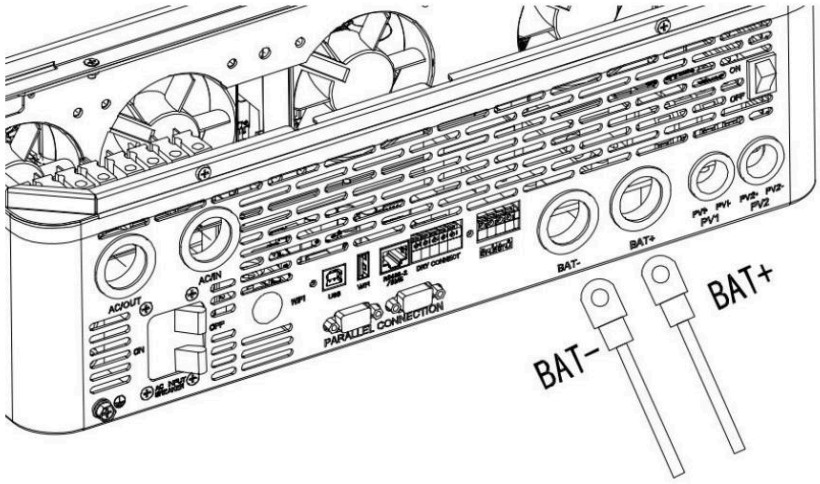
Recommended battery cable and terminal size

Models	Wire size	Rated battery discharge current	Maximum charging current	Recommended circuit breaker type	Torque value
KR-A12US	67mm²/00AWG	270A	250A	2P—300A	2-3Nm

NOTE **For a lead-acid battery, the recommended charge current is 0.2C (C = battery capacity).**

Follow these steps to connect the battery:

1. Assemble the battery ring terminal to match the recommended battery cable and terminal size.
2. Connect all battery packs as the unit requires. A battery capacity of at least 300Ah is suggested.
3. Insert the ring terminal of the battery cable flat against the battery connector of the inverter. Tighten the bolts to a torque of 2Nm. Make sure the polarity is correct at both the battery and the inverter/charger. Make sure the ring terminals are screwed tightly to the battery terminals.



Battery cables connected to BAT− and BAT+ on the inverter.

WARNING **Shock Hazard.** Install with care. Batteries connected in series have a high voltage.

CAUTION Do not place anything between the flat part of the inverter terminal and the ring terminal. This can cause overheating.

CAUTION Do not put anti-oxidant material on the terminals before the terminals are tightly connected.

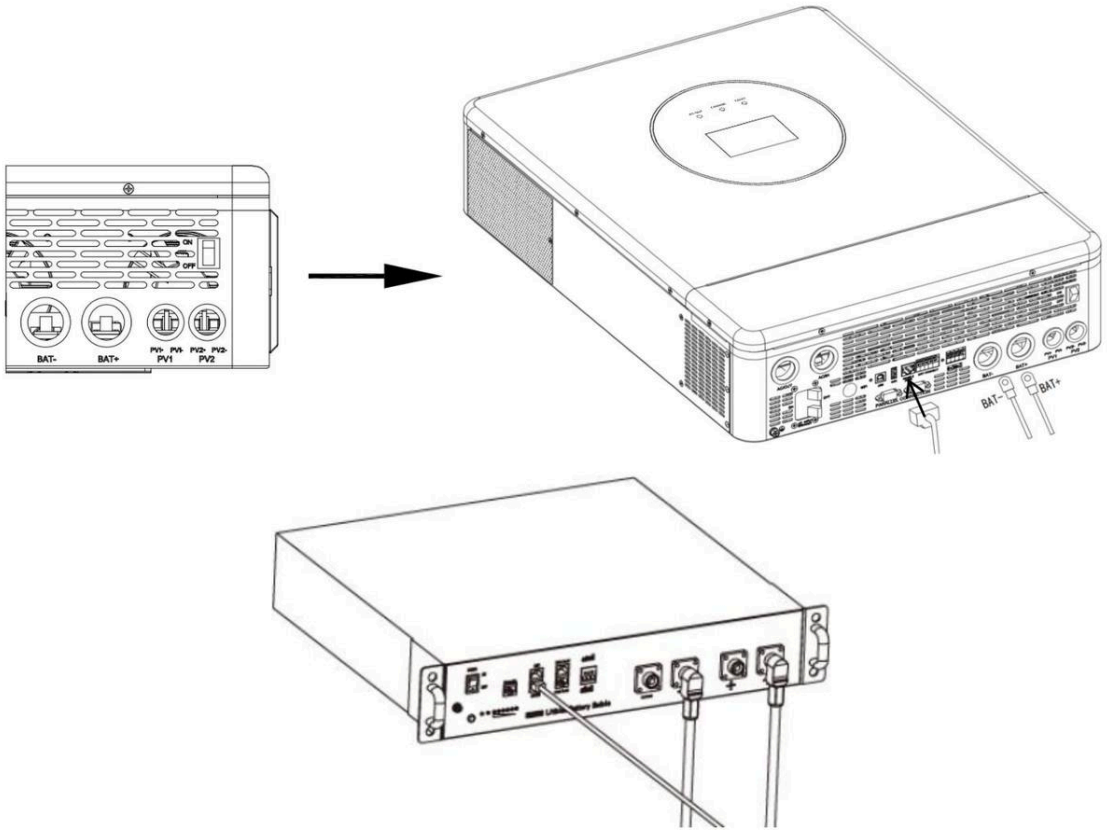
CAUTION Before you make the final DC connection or close the DC breaker/disconnect, make sure positive (+) is connected to positive (+) and negative (−) is connected to negative (−).

3.5 Lithium Battery Connection

If you use a lithium battery with this inverter, you may only use a lithium battery that KOOR has configured. The lithium battery has two connections: the RJ45 port of the BMS (battery management system) and the power cable.

Follow these steps to connect a lithium battery:

1. Assemble the battery ring terminal to match the recommended battery cable and terminal size. This is the same as for lead-acid batteries. See section 3.4, Lead-acid Battery Connection, for details.
2. Insert the ring terminal of the battery cable flat against the battery connector of the inverter. Tighten the bolts to a torque of 2-3Nm. Make sure the polarity is correct at both the battery and the inverter/charger. Make sure the ring terminals are screwed tightly to the battery terminals.
3. Plug one end of the battery's RJ45 cable into the BMS communication port (RS485-2/BMS) of the inverter.
4. Plug the other end of the RJ45 cable into the battery communication port (RS485).



Lithium battery connection: power cables to BAT−/BAT+ and the RJ45 cable to the RS485-2/BMS port.

> **NOTE** If you use a lithium battery, make sure the BMS communication cable is connected between the battery and the inverter. See section 3.7 for other settings.

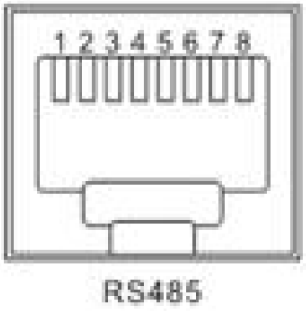
3.6 Lithium Battery Communication and Setting

> Using a KOOR battery? **Appendix A** gives the exact port, addresses and settings for the KOOR 16K Pro and the KOOR Active Stack.

Plug the battery end of the RJ45 cable into the BMS communication port of the inverter.

Make sure the lithium battery BMS port connects to the inverter pin-to-pin. The pin assignment of the inverter BMS (RS485-2/BMS) port is shown below:

No.	BMS port (RS485-2/BMS)
1	RS485-B
2	RS485-A
3	--
4	CAN-H
5	CAN-L
6	--
7	RS485-A
8	RS485-B



RS485 port, pins 1 to 8.

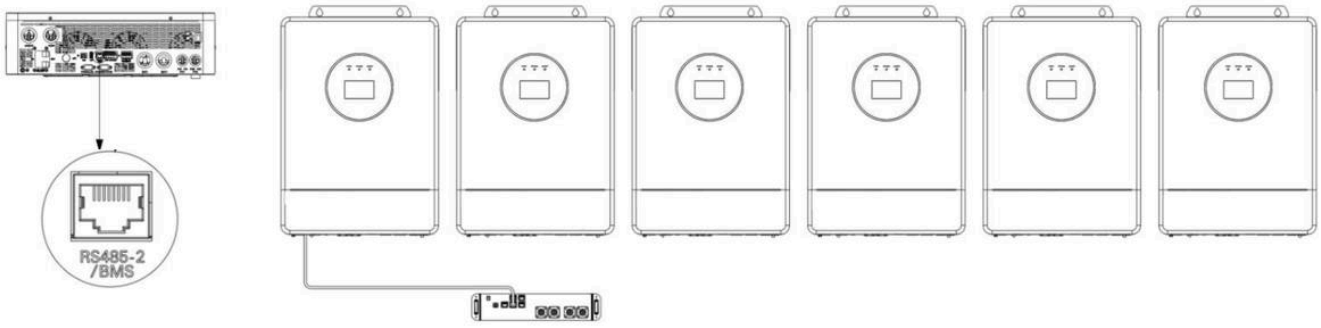
3.7 LCD Setting

When you connect a lithium battery BMS, match the battery list parameters in section 6 to the battery pack specifications. In the inverter settings, set the battery type to the closest one of "LFP15", "LFP16" or "NCM14". After you set the battery type in the setting named "Battery type", go to the setting named "RS485-2" to choose the communication mode. Last, go to "BMS protocol" to choose the matching BMS communication protocol.

Setting	Option	Description
Battery type	USER	User-defined. All battery parameters can be set.
	SLD	Sealed lead-acid battery. Constant-voltage charge voltage: 57.6V. Float charge voltage: 55.2V.
	FLD	Vented (flooded) lead-acid battery. Constant-voltage charge voltage: 58.4V. Float charge voltage: 55.2V.
	GEL **(default)**	Colloidal (gel) lead-acid battery. Constant-voltage charge voltage: 56.8V. Float charge voltage: 55.2V.
	LFP15/LFP16	Lithium iron phosphate (LFP) battery. LFP15 and LFP16 mean 15 and 16 cells in series. For 16 in series, the default constant-voltage charge voltage is 56.8V. For 15 in series, the default constant-voltage charge voltage is 53.2V. These can be adjusted.
	NCM14	Ternary lithium (NCM) battery. Adjustable. The default constant-voltage charge voltage for 14 in series (NCM 14) is 57.6V.
RS485-2 communication	DIS **(default)**	Turns off BMS communication on RS485-2. The PC and remote monitoring protocol can still be used.
	BMS	Uses the RS485-2 port for BMS communication.
BMS protocol	When "RS485-2 communication" is set to "BMS", choose the matching lithium battery maker's protocol.	
	Pylon, Aoguan, Oulite, Gotion, Sunwoda, CF, Dyness, Pace, BST, Fox-ess, AEC, PylonV3.5, Pylon Can, EVI, Voltronic.	

3.8 Communicating with the Battery BMS in a Parallel System

To use BMS communication in a parallel system, connect the BMS communication cable between the battery and one inverter of the parallel system. Connecting it to the master inverter of the parallel system is recommended.



In a parallel system, the battery BMS cable goes to the RS485-2/BMS port of one inverter.

3.9 AC Input/Output Connection

CAUTION Before connecting to the AC input power source, install a separate AC breaker between the inverter and the AC input power source. This lets you safely disconnect the inverter during maintenance and fully protects it from AC input over-current.

CAUTION There are two terminal blocks, marked "IN" and "OUT". Do NOT mix up the input and output connectors.

WARNING All wiring must be done by qualified personnel.

WARNING Using the right cable for the AC input connection is very important for system safety and efficient operation. To reduce the risk of injury, use the recommended cable size shown below.

Suggested cable requirements for AC wires

Models	Recommended AC input wire size	Maximum bypass input current	Recommended circuit breaker type	Torque value
KR-A12US	13mm²/6AWG	60A	3P—63A	1.2-1.6Nm

Models	Recommended AC output wire size	Maximum bypass output current	Recommended circuit breaker type	Torque value
KR-A12US	13mm²/6AWG	60A	3P—63A	1.2-1.6Nm

Follow these steps to connect AC input and output:

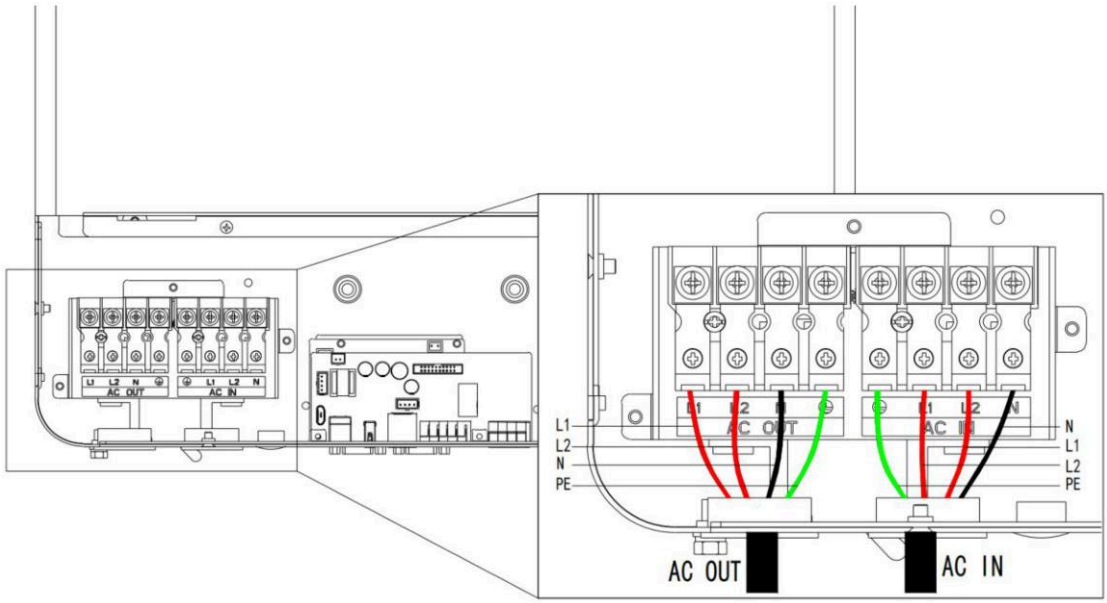
1. Before you make the AC input/output connection, open (turn off) the DC protector or disconnector first.
2. Strip 10 mm (0.39 in) of insulation from each of the eight conductors. Shorten the phase conductor L and the neutral conductor N by 3 mm (0.12 in).
3. Insert the AC input wires as marked by the polarities on the terminal block, and tighten the terminal screws. Be sure to connect the PE (protective earth, ground) conductor first.

PE (ground symbol)	→ Ground (yellow-green)
L1, L2	→ LINE

N	→ Neutral

WARNING Make sure the AC power source is disconnected before you hardwire it to the unit.

4. Next, insert the AC output wires as marked by the polarities on the terminal block, and tighten the terminal screws. Be sure to connect the PE (protective earth, ground) conductor first.
5. Make sure the wires are securely connected.



AC OUT and AC IN terminal blocks (L1, L2, N, PE).

CAUTION **Important:** Connect the AC wires with the correct polarity. If the L and N wires are reversed, it may cause a grid/utility (mains) short circuit when inverters are running in parallel.

CAUTION Appliances such as air conditioners need at least 2~3 minutes before they restart, because the refrigerant gas inside the circuits needs time to balance. If power is lost and comes back quickly, it will damage your connected appliances. To prevent this damage, check with the air conditioner manufacturer before installation to see if it has a time-delay function. If it does not, this inverter will trigger an overload fault and cut off output to protect your appliance, but sometimes the air conditioner can still be damaged inside.

3.10 PV Connection

CAUTION Before connecting to the PV (solar) modules, install a separate DC circuit breaker between the inverter and the PV modules.

WARNING All wiring must be done by qualified personnel.

WARNING Do not connect one PV string to more than one inverter at the same time. This may damage the inverter.

WARNING Using the right cable for the PV module connection is very important for system safety and efficient operation. To reduce the risk of injury, use the recommended cable size shown below.

Models	Recommended PV wire size	Maximum PV input current	Recommended circuit breaker type	Torque value
KR-A12US	5.2mm²/10AWG	27A	2P—32A	1.2-1.6Nm

PV Module Selection

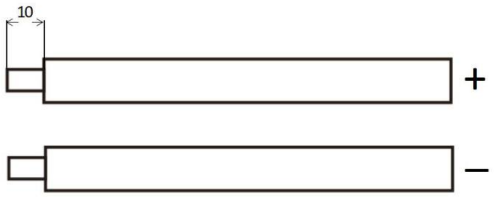
When you choose PV modules, check these points:

1. The open circuit voltage (Voc) of the PV array must not be higher than the inverter's maximum PV array open-circuit voltage.
2. The open circuit voltage (Voc) of the PV array should be higher than the start-up voltage.

INVERTER MODEL	KR-A12US
Max. PV Array Open Circuit Voltage	500Vdc
Start-up Voltage	120Vdc
PV Array MPPT Voltage Range	90Vdc~450Vdc

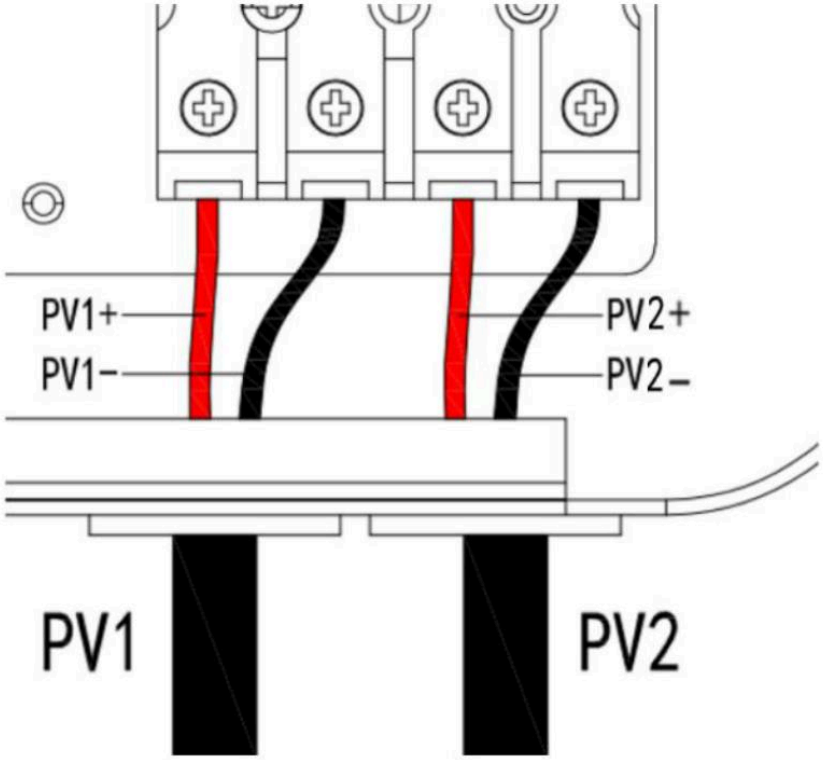
Follow these steps to connect the PV modules:

1. Strip 10 mm (0.39 in) of insulation from the positive and negative conductors.



Strip 10 mm from the + and – conductors.

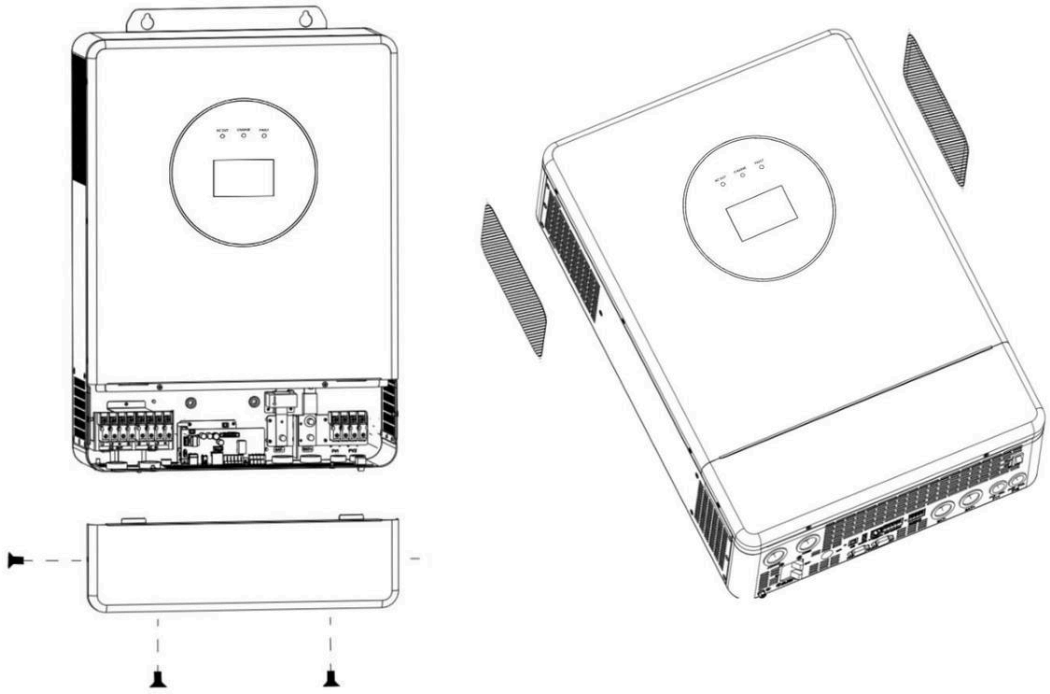
2. Check that the polarity of the cable from the PV modules and the PV input connectors is correct. Then connect the positive pole (+) of the cable to the positive pole (+) of the PV input connector. Connect the negative pole (–) of the cable to the negative pole (–) of the PV input connector.
3. Make sure the wires are securely connected.



PV terminals: PV1+, PV1−, PV2+, PV2−.

3.11 Final Assembly

After all wiring is connected, put the bottom (base) cover and the dust covers back on as shown in the diagram below:



Reinstall the bottom cover with its screws, and fit the dust covers on both sides.

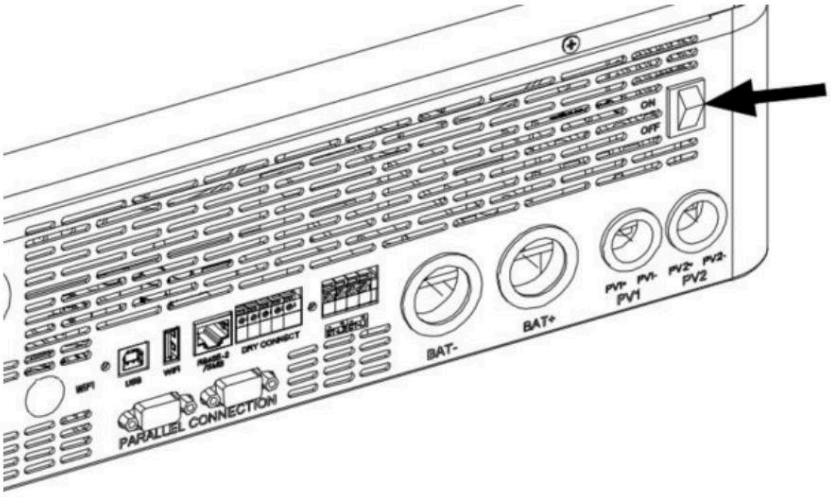
3.12 Starting the Inverter

1. Close (turn on) the battery circuit breaker.
2. Press the rocker switch at the bottom of the inverter. The screen and the indicator light up. This shows the inverter is now on.
3. Close the circuit breakers for PV, AC input and AC output, in that order.
4. Turn on the loads one at a time. Start with the smallest load and work up to the largest.

4. Operation

4.1 Power ON/OFF

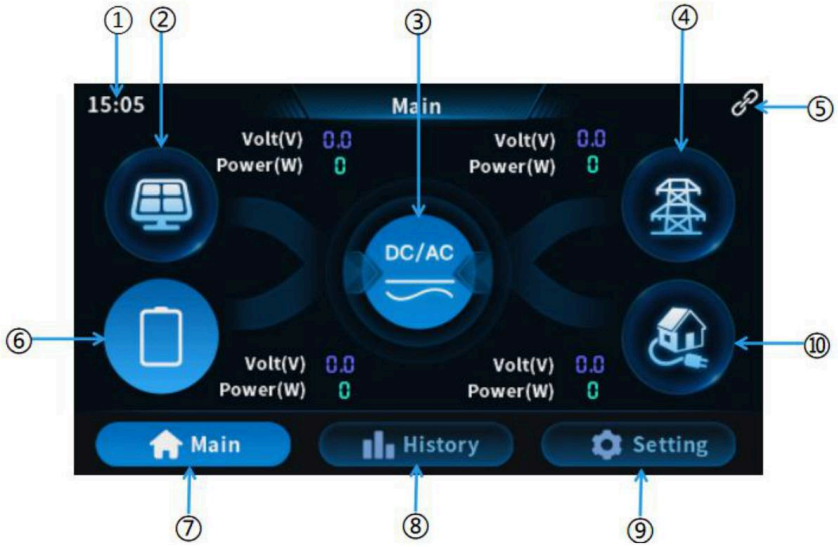
First make sure the unit is installed correctly and the batteries are connected well. Then press the On/Off switch to turn on the unit. The switch is on the bottom of the case (see the arrow below).



On/Off switch location (arrow) on the bottom of the case

4.2 Operation and Display Panel

The operation and display panel is on the front of the inverter (shown below). It has three indicators and an LCD display. They show the operating status and the input and output power information.



LCD main screen

No.	Item	No.	Item
①	Time	⑥	Battery parameters
②	Photovoltaic (PV) parameters	⑦	Main
③	System information	⑧	Historical data
④	AC input parameters	⑨	Setting
⑤	Screen communication	⑩	AC output parameters

LCD screen setting menu



"Setting" menu: Normal, Battery, Parallel, BMS, Hybrid and Others

How to view real-time data

On the LCD main screen, press the screen to scroll through the machine's real-time data.

4.3 Setup Parameters Description

How to use the screen:

1. Press the "Setting" button to open the setup menu.
2. In the Settings menu, pick the option you want to set and tap it to open it.
3. Press "<" and ">" to select the value you want.
4. Press "Confirm" to save the new setting.

> **NOTE** In parallel mode, before startup, all machines copy the settings of the host. The host is the machine whose screen shows "Main". After startup, a setting changed on any machine is copied to all the other machines in the system.

Parameter name	Settings	Description
Normal setting		
AC output mode priority	Mains **default**	Mains (grid) priority mode. The unit switches to the inverter only when the mains fails.
	Battery	Battery priority mode. The unit switches to mains power only when the battery is under voltage or below the setting of parameter **"Battery Power to mains"**. It switches back to battery discharge only when the battery is fully charged or above the setting of parameter **"Mains power to battery"**.
	PV	Solar priority mode. The unit switches to the mains when solar fails or when the battery is below the setting of parameter **"Battery Power to mains"**.
	Hybrid	Hybrid mode. You set this mode with the **"Hybrid way"** setting (Hybrid setting menu).
AC output phase voltage setting	120Vac **default**	Can be set to 100Vac / 105Vac / 110Vac / 120Vac / 127Vac. Default is 120V. The rated output power is reduced to: (Power Rate) × (Vset / 120).
AC output Frequency	50.0Hz	Bypass self-adaptation: when the mains is connected, the unit automatically matches the mains frequency. When the mains is disconnected, you set the output frequency in this menu. For 120Vac, the default output frequency is 60Hz.
	60.0Hz **default**	
AC Input Voltage Range	UPS **default**	Narrow mains input voltage range for the 120Vac machine: 90~140Vac. (The lower input voltage limit becomes 85V when the system voltage is 100V / 105V.) Frequency range: 47~55Hz (50Hz); 57Hz~65Hz (60Hz).
	Generator	Use this mode for diesel generator input. In this mode: Narrow AC input voltage range for the 120Vac machine: 90~140Vac. Frequency range: 40~70Hz.
Power saving mode	Disable **default**	Power saving mode is off.
	Enable	With power saving mode on: if the load on each phase is zero or less than 25W, the inverter output turns off after a set delay time. When the load on any phase is more than 50W, the inverter restarts automatically.
Charger priority (cannot be set in Hybrid mode)	Hybrid charging **default**	PV and mains hybrid charging. PV charging comes first. When PV energy is not enough, mains charging makes up the difference. When PV energy is enough, mains charging stops. Note: PV charging and mains charging can run at the same time only when the load is running on mains bypass output. When the inverter is working, only PV charging can start.
	Mains priority	Mains priority charging. PV charging starts only when mains charging fails.
	PV priority	PV priority charging. Mains charging starts only when PV charging fails.
	PV only charging	PV charging only. Mains charging is not turned on.

Parameter name	Settings	Description
AC output phase setting (Turn off the chassis power switch before setting)	Same phase	Single-phase mode. The phase angle between L1 and L2 is 0 degrees; L1 and L2 are connected in parallel. With the AC output voltage set to 120Vac, L1-N and L2-N voltage is 120Vac, and L1-L2 voltage is 0Vac. This can be set only when the output is switched off.
	Split 120 P-seq	Split-phase mode. The phase difference between L1 and L2 is positive 120 degrees; L1 and L2 are in split connection. With the AC output voltage set to 120Vac, L1-N and L2-N voltage is 120Vac, and L1-L2 voltage is 208Vac. This can be set only when the output is switched off.
	Split 180 **default**	Split-phase mode. The phase difference between L1 and L2 is 180 degrees; L1 and L2 are in split connection. With the AC output voltage set to 120Vac, L1-N and L2-N voltage is 120Vac, and L1-L2 voltage is 240Vac. This can be set only when the output is switched off.
	Split 120 N-seq	Split-phase mode. The phase difference between L1 and L2 is negative 120 degrees; L1 and L2 are in split connection. With the AC output voltage set to 120Vac, L1-N and L2-N voltage is 120Vac, and L1-L2 voltage is 208Vac. This can be set only when the output is switched off.
Max charging current	60A **default**	Maximum total charging current setting. Setting range: 0~250A
Max AC charger current	60A **default**	Setting range: 0~160A
Buzzer Mute	Disable **default**	The alarm beeps when the status of the main input source changes or when the inverter has a fault.
	Enable	The buzzer is silent.
Fully charging judgment current setting	3A **default**	Lead-acid battery: when the battery voltage is equal to or above the floating charge value, and the charging current is below this setting, the battery is treated as fully charged and charging stops. Lithium battery: when the battery voltage is equal to or above the constant voltage charging value, and the charging current is below this setting, the battery is treated as fully charged and charging stops.
N-PE connection switchover is enabled	Disable **default**	Never connect the N (neutral) wire to the PE (ground) wire automatically, under any working condition.
	Enable	When the unit is off-grid with no mains input, the neutral line automatically connects to PE. When the bypass has mains input, the neutral line automatically disconnects from PE.
AC Overload restart Enabled	Disable	Automatic restart after overload is not allowed. If the output shuts down from overload, the machine will not start up again.
	Enable **default**	Automatic restart after overload is allowed. If the output shuts down from overload, the machine restarts the output after a 3-minute delay. After a total of 5 restarts within 30 minutes, the system no longer restarts.
Battery setting		
Automatic battery activation	Disable	When the battery is dormant (asleep) or not connected, PV or mains does **not** automatically wake the battery to turn on the battery output.
	Enable **default**	When the battery is dormant (asleep) or not connected, PV or mains automatically wakes the battery to turn on the battery output.

Parameter name	Settings	Description
Battery type	USER	User-defined. All battery parameters can be set.
	SLD	Sealed lead-acid battery. Constant-voltage charge voltage: 57.6V. Floating charge voltage: 55.2V.
	FLD	Vented (flooded) lead-acid battery. Constant-voltage charge voltage: 58.4V. Floating charge voltage: 55.2V.
	GEL **default**	Colloidal (gel) lead-acid battery. Constant-voltage charge voltage: 56.8V. Floating charge voltage: 55.2V.
	LFP15/ LFP16	Lithium iron phosphate (LiFePO4) battery, LFP15/LFP16, for 15-string and 16-string packs. For 16 strings, the default constant-voltage charge voltage is 56.8V. For 15 strings, the default constant-voltage charge voltage is 53.2V. Both can be adjusted.
	NCM14	Ternary lithium battery; adjustable. For the NCM 14-series pack, the default constant voltage charging voltage is 57.6V.
Battery pack rate AH	300AH	Lead-acid battery capacity.
Battery constant charge voltage	57.6V **default**	C.V. (constant voltage) charging voltage setting. Setting range: 48V~58.4V, in steps of 0.1V. Valid for user-defined battery and lithium battery.
Battery floating charge voltage	57.6V **default**	Floating charge voltage. Setting range: 48V~58.4V, step: 0.1V.
Battery recharge recovery point	53V **default**	After the battery is fully charged, the inverter stops charging. It starts charging again when the battery voltage drops below this value.
Battery Power to mains	47.2V **default**	When parameter **"AC output mode priority"** is set to "Battery" and the battery voltage is below this setting, the output switches from the inverter to the mains. Setting range: 44V~52V.
Mains power to battery	57.6V **default**	When parameter **"AC output mode priority"** is set to "Battery" and the battery voltage is above this setting, the output switches from the mains to the inverter. Setting range: 48V~60V.
Low battery voltage alarm	47.2V **default**	Battery under-voltage alarm point. When the battery voltage is below this point, an under-voltage alarm is given (01 fault). The output is **not** turned off. Setting range: 40V~52V, in steps of 0.1V.
Delay off discharge	46.4V **default**	Over-discharge voltage. When the battery voltage is below this point, the unit waits for the time set in parameter **"Off discharge delay time"**, then turns off the inverter output. Setting range: 40V~48V, in steps of 0.1V.
Battery over discharge voltage (immediately off discharge)	44.4V **default**	Battery discharge limit voltage. When the battery voltage is below this point, the output turns off right away (02 fault). Setting range: 40V~52V, in steps of 0.1V. Valid for user-defined battery and lithium battery.
Off discharge delay time	30s **default**	Over-discharge delay time. When the battery voltage is below parameter **"Delay off discharge"**, the inverter output turns off after the delay set here. Setting range: 5s~120s, in steps of 1s.
Over discharge recovery (02 fault)	51.2V **default**	After over-discharge protection turns off the inverter output, the battery voltage must rise above this value before the inverter AC output comes back on.

Parameter name	Settings	Description
Timed charging	Disable **default**	This function is off.
	Enable	This function is on. The utility charges the battery and carries the load only during the charging time slots you set, or when the battery is under voltage.
	0h 0min	1st slot charging start; range: 00:00-23:59
	0h 0min	1st slot charging end; range: 00:00-23:59
	0h 0min	2nd slot charging start; range: 00:00-23:59
	0h 0min	2nd slot charging end; range: 00:00-23:59
	0h 0min	3rd slot charging start; range: 00:00-23:59
	0h 0min	3rd slot charging end; range: 00:00-23:59
Timed discharging	Disable **default**	This function is off.
	Enable	This function is on. The battery discharges only during the discharging time slots you set, or when the utility is not available.
	0h 0min	1st slot discharging start; range: 00:00-23:59
	0h 0min	1st slot discharging end; range: 00:00-23:59
	0h 0min	2nd slot discharging start; range: 00:00-23:59
	0h 0min	2nd slot discharging end; range: 00:00-23:59
	0h 0min	3rd slot discharging start; range: 00:00-23:59
	0h 0min	3rd slot discharging end; range: 00:00-23:59
Constant voltage charging time	120min **default**	C.V. charging maximum time setting. This is the longest time the unit charges at the voltage set in parameter **"Battery constant charge voltage"** during constant-voltage charging. Setting range: 5min~900min, in steps of 1 minute. Valid for user-defined battery and lithium battery.
Parallel setting		
Parallel AC output mode setting (Turn off the rocker switch before configuration)	Single **default**	When one inverter is used alone, the default is SIG mode.
	PAL	Parallel operation in single phase. See **8.3** Wiring Diagram.
Communication address	1 **default**	In parallel mode, set the ID in the range 1-6. Each ID must be different. The IDs are assigned automatically when power is first turned on. Single mode setting range: 1-254.
Battery connected independently	Disable **default**	All battery packs are used in parallel (shared).
	Enable	Battery packs are not connected in parallel. Each battery is connected to its own inverter.

Parameter name	Settings	Description
BMS setting		
Communication type	BMS OFF **default**	BMS communication is turned off.
	485-BMS	The BMS communicates through RS485-2.
	CAN-BMS	The BMS communicates through CAN.
BMS protocol	When BMS communication is turned on, select the matching lithium battery manufacturer brand for communication.	
	Pylon, Aoguan, Oulite, Gotion, Sunwoda, CF, Dyness, Pace, BST, Fox-ess, AEC, PylonV3.5, Pylon Can, EVI, Voltronic.	
SOC discharging alarm (Optional when BMS is enabled)	15% **default**	With the **BMS** setting item enabled: the machine gives alarm 07 when the battery capacity rate (SOC) is below this value. The alarm clears when the battery capacity rate is higher than 5% of this value.
SOC switching mains (optional when BMS is enabled)	10% **default**	With the **BMS** setting item enabled: the machine switches to the mains when the battery capacity rate is below this value and mains power is available.
SOC switching inverter (optional when BMS is enabled)	100% **default**	With the **BMS** setting item enabled, this works in battery priority mode. When the battery capacity rate is above this value, the machine switches from the mains to inverter mode.
SOC discharge off (optional when BMS is enabled)	5% **default**	With the **BMS** setting item enabled: when the battery capacity rate is below this value, the machine reports fault 08 and cuts off the power supply or output.
SOC charging off (optional when BMS is enabled)	100% **default**	With the **BMS** setting item enabled: when the battery capacity rate is above this value, the inverter treats the battery as full. If battery priority mode is in use, the unit switches the load from the mains back to the battery.
SOC exit forced charging (optional when BMS is enabled)	10% **default**	With the **BMS** setting item enabled: when the inverter gets a forced-charging command from the BMS, it force-charges the battery. When the battery capacity rate rises above this value, the inverter leaves forced-charging mode and goes on running in the current set mode.
Charging strategy options	BMS+Inverter	The battery's maximum charging current is limited by the machine's descending logic.
	BMS only	The battery's maximum charging current is limited by the BMS current limit value.
	Manual setup	The battery's maximum charging current is limited by the value set in the setting item.

Parameter name	Settings	Description
Hybrid setting		
Hybrid way (available only in Hybrid mode)	Energy To Output **default**	Energy made by the inverter goes only to the load (generation power < load power). Note: mixed load — the mains and PV carry the load together; when PV is not enough, the mains makes up the difference.
	Curr Anti-Rev (CT)	Anti-countercurrent (anti-backfeed). Self-use based on the load power; no current is sent into the mains.
Zero-Export Power	50W **default**	Based on the power set by the meter. Setting range: ±500.
Hybrid Max.Bat Curr	272A **default**	Works in Hybrid mode. Limits the battery's maximum discharge current.
Hybrid priority	Disable **default**	PV energy powers the load first. Extra energy goes to the grid.
	Enable	PV energy goes to the power grid first. Extra energy then goes to the load.
External CT Get Host Enable	Disable **default**	Controls only the grid-connected power of this inverter.
	Enable	Anti-backflow host. Plug the CT into this inverter to control the total grid-connection power.
Others setting		
Backlight off	5min **default**	Sets how long the screen stays lit (screen brightness time).
Brightness	10% **default**	Sets the screen brightness.
On/OFF control	Start up	Turns on the inverter AC output.
	Shut down	Turns off the inverter AC output.
Restart control	Restart	Restarts the inverter.
Language	English **default**	Switches between Chinese / English / español.
Present time	2000/01/01 0:0:00	Sets the time.
Screen reset	Reset	Resets the screen position.
Screen buzzer	Disable	The screen buzzer is silent.
	Enable **default**	The buzzer sounds when you touch the screen.

5. Operating Mode Description

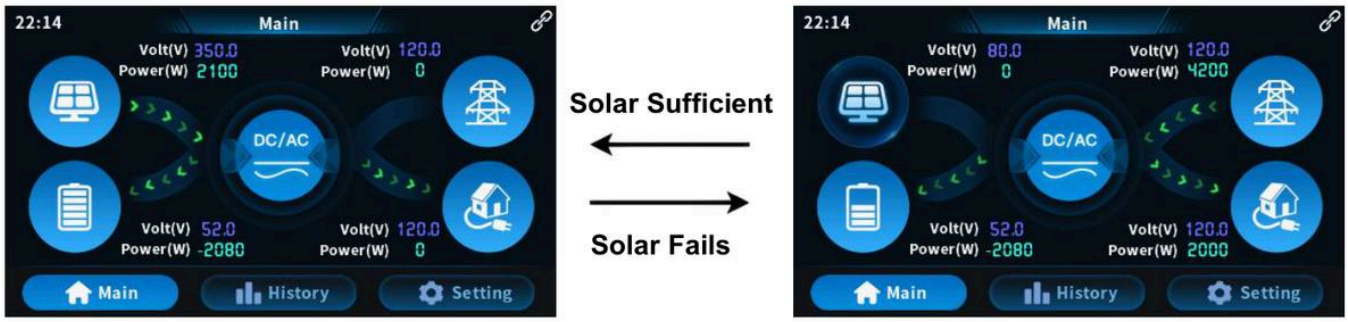
5.1 Charging Mode

1) PV priority

The PV (solar) modules charge the battery first. The grid (mains) charges the battery only when the PV system fails.

During the day, solar power is fully used for charging. At night, charging switches to the grid (mains).

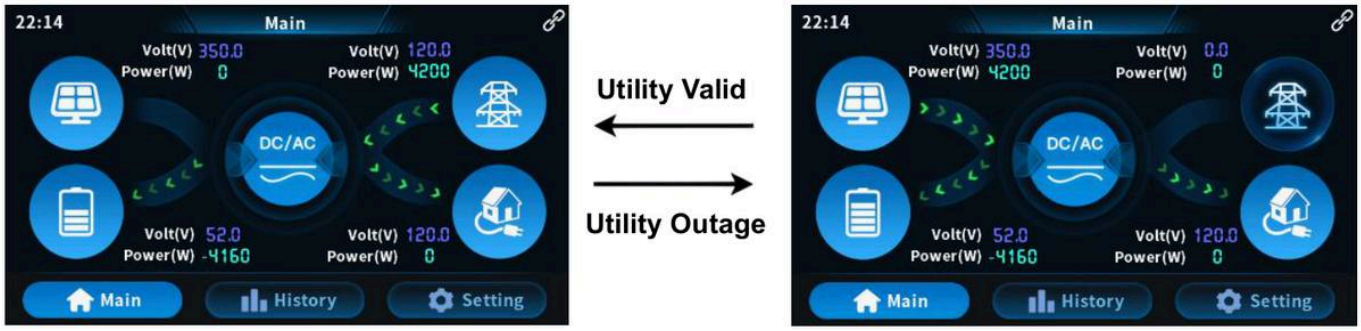
This keeps the battery level up. It is ideal for areas where the grid is fairly stable and electricity is fairly expensive.



PV priority charging: the unit moves between the two states as solar is sufficient or solar fails.

2) Mains priority

The grid (mains) charges the battery first. PV charging can start only when the grid (mains) fails.

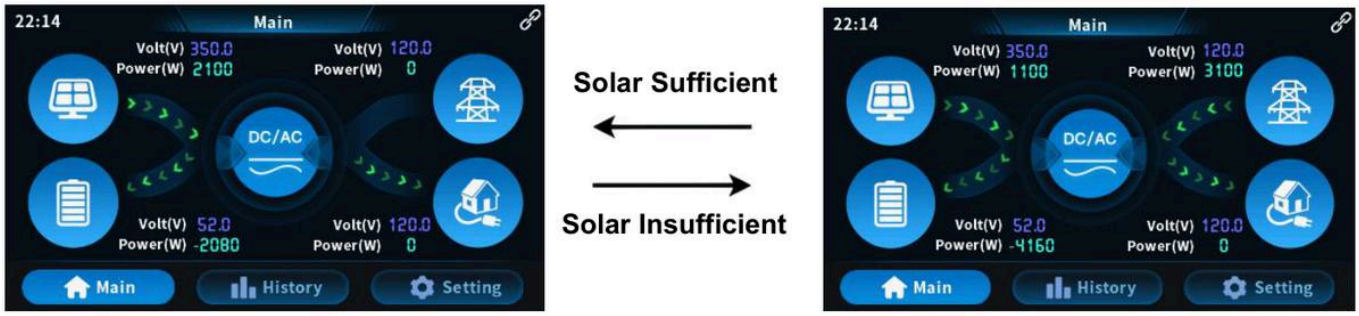


Mains priority charging: the unit moves between the two states as the utility is valid or the utility has an outage.

3) Hybrid charging

PV and the grid (mains) charge the battery together. PV MPPT charging comes first. When PV energy is not enough, the grid (mains) makes up the difference. When PV energy is enough again, the grid (mains) stops charging.

This is the fastest charging mode. It suits areas where the power grid is unstable, because it keeps enough backup power ready at all times.



Hybrid charging: the unit moves between the two states as solar is sufficient or solar is insufficient.

4) Only Solar

Only PV charges the battery. The grid (mains) does not charge it.

This is the most energy-efficient way, because only the solar panels charge the battery. It is usually used in areas with good sunlight.



Only Solar charging.

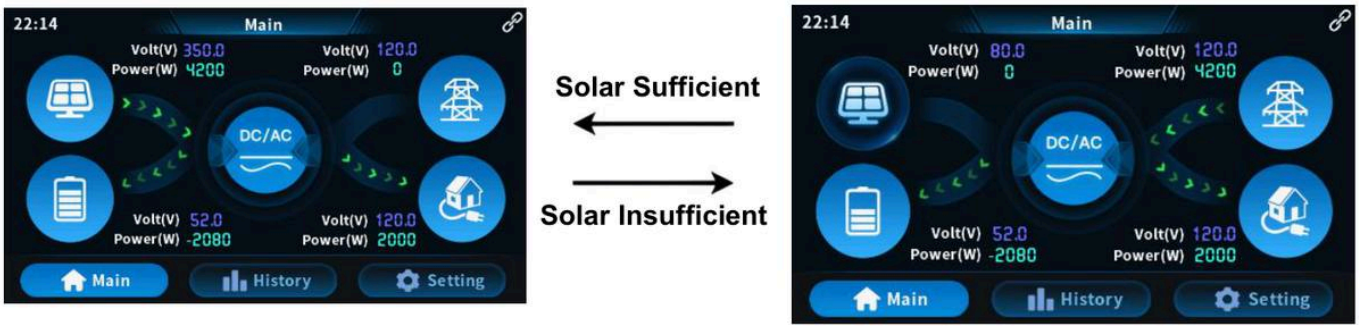
5.2 AC Output Mode

PV Priority Mode

PV (solar) and the batteries power the load. The unit switches to the grid (mains) when PV charging fails.

This mode makes the most of solar energy and keeps the battery charged. It suits areas where the grid is fairly stable.

- **Power supply priority: Solar → Battery → Utility** (Solar Sufficient)
- **Power supply priority: Solar → Utility → Battery** (Solar Insufficient)



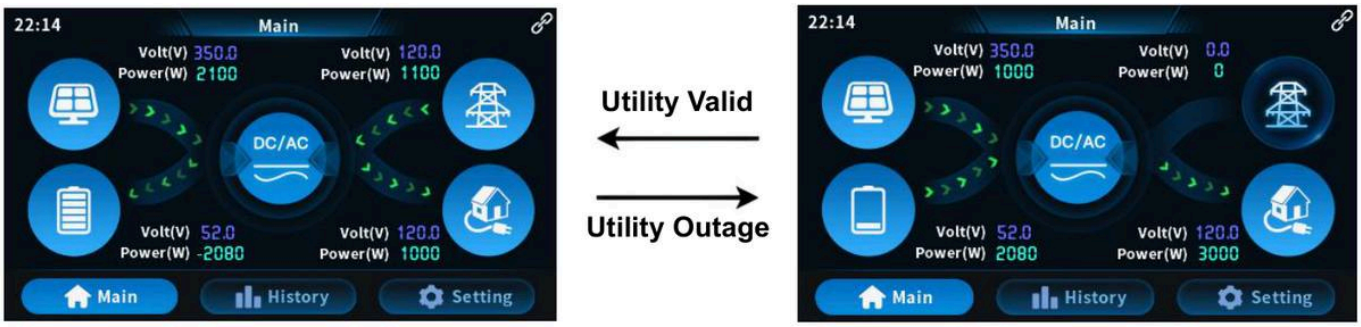
PV priority output: the unit moves between the two states as solar is sufficient or solar is insufficient.

Mains Priority Mode

The unit switches to the inverter only when the grid (mains) fails. While grid (mains) power is present, the unit uses the grid (mains) for both charging and powering the load.

In this mode the unit works like a backup UPS. It suits areas where the grid is unstable. Switching does not affect PV charging.

- **Power supply priority: Utility → Solar → Battery**



Mains priority output: the unit moves between the two states as the utility is valid or the utility has an outage.

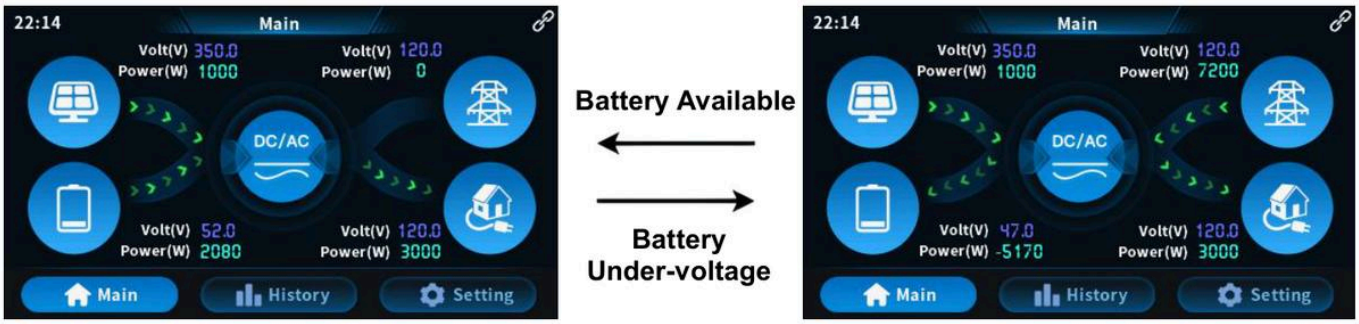
Battery Priority Mode

The unit switches to the grid (mains) only when the battery voltage, while discharging, drops below the setting of the parameter **"Battery Power to mains"**.

When the battery, while charging, rises above the setting of the parameter **"Mains power to battery"**, the unit switches back to battery discharge mode.

This lets the battery cycle between charging and discharging. This mode makes the most use of DC power and is used in areas where the grid is stable. Switching does not affect PV charging.

- **Power supply priority: Solar → Battery → Utility**



Battery priority output: the unit moves between the two states as the battery is available or the battery is under-voltage.

Hybrid Mode

- **Energy To Output:** The inverter sends the energy it generates only to the load (generation power < load power). **Note:** Mixed load — the grid (mains) and PV carry the load together; when PV is not enough, the grid (mains) makes up the difference.
- **Curr Anti-Rev (CT):** Anti-counter current (anti-backfeed). The unit powers the house for self-use based on the load power and does not send current into the grid (mains).

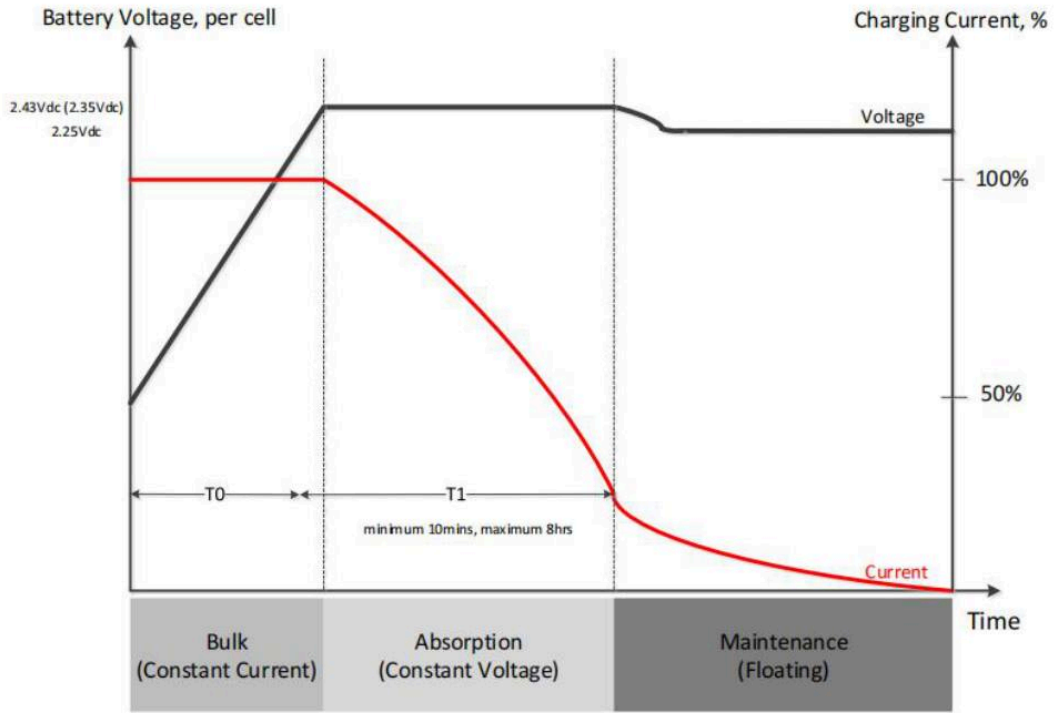


Hybrid mode.

6. Battery Type Parameters

6.1 Lead-acid Battery

Parameter \ Battery type	Sealed lead-acid battery (SLD)	Colloidal lead-acid battery (GEL)	Vented lead-acid battery (FLD)	User-defined (User)
Over voltage disconnection voltage	60V	60V	60V	40~60V (Adjustable)
Battery fully charged recovery point	52V (Adjustable)	52V (Adjustable)	52V (Adjustable)	52V (Adjustable)
Boost charge voltage	57.6V	56.8V	58.4V	40~60V (Adjustable)
Floating charge voltage	55.2V	55.2V	55.2V	40~60V (Adjustable)
Under voltage alarm voltage (01 fault)	44V	44V	44V	40~60V (Adjustable)
Under voltage alarm voltage recovery point (01 fault)	Under voltage alarm voltage + 0.8V			
Low voltage disconnection voltage (02 fault)	42V	42V	42V	40~60V (Adjustable)
Low voltage disconnection voltage recovery point (02 fault)	52V (Adjustable)	52V (Adjustable)	52V (Adjustable)	52V (Adjustable)
Discharge limit voltage	40V	40V	40V	40~60V (Adjustable)
Over-discharge delay time	5s	5s	5s	1~30s (Adjustable)
Boost charge duration	120 minutes	120 minutes	120 minutes	10~600 minutes (Adjustable)
Lead-acid charging curve	See the chart below.			



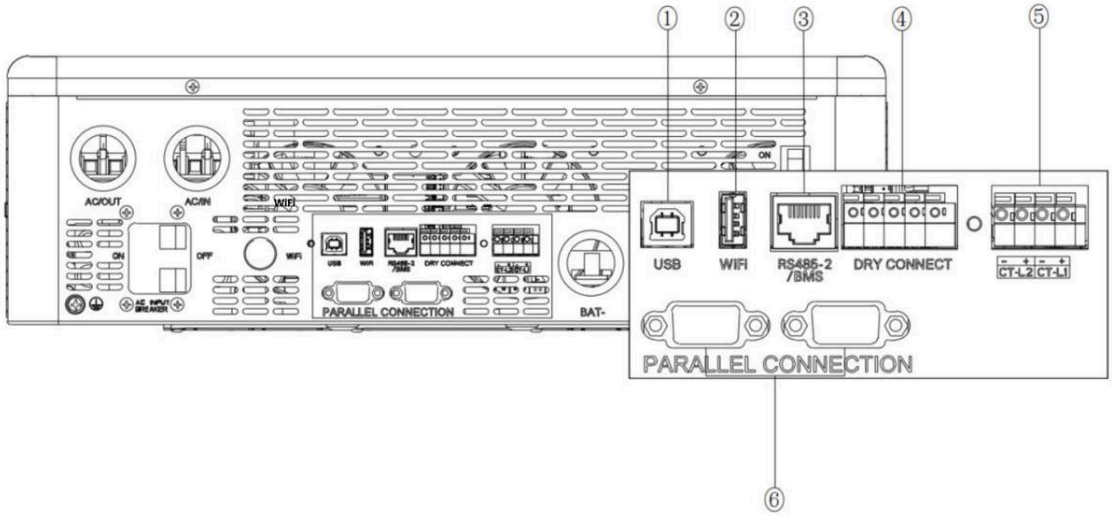
Lead-acid charging curve: battery voltage per cell (2.43Vdc (2.35Vdc) and 2.25Vdc marks) and charging current (%) over time. Stages: Bulk (Constant Current) for time T0; Absorption (Constant Voltage) for time T1 (minimum 10 mins, maximum 8 hrs); then Maintenance (Floating).

6.2 Lithium Battery

Parameter \ Battery type	Ternary lithium battery (NCM14)	Lithium iron phosphate battery (LFP16)	Lithium iron phosphate battery (LFP15)
Over voltage disconnection voltage	60V	60V	60V
Battery fully charged recovery point	54.6V (Adjustable)	53.0V (Adjustable)	50.4V (Adjustable)
Boost charge voltage	57.4V (Adjustable)	57.6V (Adjustable)	53.2V (Adjustable)
Floating charge voltage	57.4V (Adjustable)	57.6V (Adjustable)	53.2V (Adjustable)
Under voltage alarm voltage (01 fault)	47.6V (Adjustable)	47.2V (Adjustable)	46.4V (Adjustable)
Under voltage alarm voltage recovery point (01 fault)	Under voltage alarm voltage + 0.8V		
Low voltage disconnection voltage (02 fault)	44V (Adjustable)	46.4V (Adjustable)	45.6V (Adjustable)
Low voltage disconnection voltage recovery point (04 fault)	49.2V (Adjustable)	51.2V (Adjustable)	49.6V (Adjustable)
Discharge limit voltage	43.4V (Adjustable)	44.4V (Adjustable)	43.6V (Adjustable)
Over-discharge delay time	30s (Adjustable)	30s (Adjustable)	30s (Adjustable)
Boost charge duration	120 minutes (Adjustable)	120 minutes (Adjustable)	120 minutes (Adjustable)

7. Communication

7.1 Overview



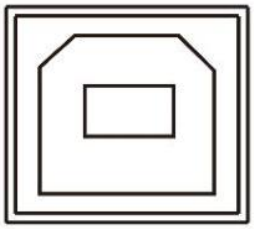
Communication ports on the bottom of the case (callouts 1–6).

No.	Port	No.	Port	No.	Port
1	USB-B port	2	WiFi port	3	RS485-2/BMS port
4	Dry contact port	5	CT sampling detection (anti feed-in)	6	Parallel communication port

7.2 USB Communication Port

This is a USB communication port. Use it for USB communication with the optional PC host software.

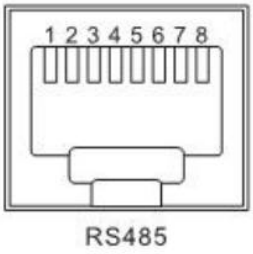
To use this port, install the matching **"USB to serial chip CH340T driver"** on the computer.



USB-B port

7.3 RS485 Communication Port

There is one RS485 port: RS485-2. The RS485-2 port talks directly with the lithium battery BMS over RS485 (custom).



RS485 port, pins 1–8

RS485-2 pinout:

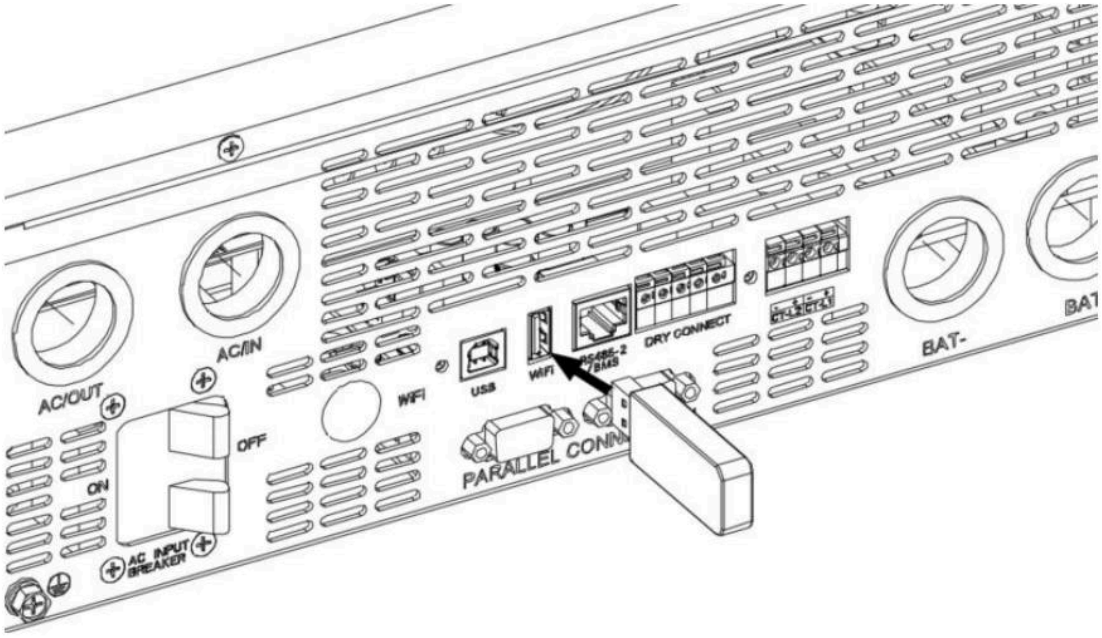
- Pin 1: RS485-B
- Pin 2: RS485-A
- Pin 4: CAN-H
- Pin 5: CAN-L
- Pin 7: RS485-A
- Pin 8: RS485-B
- Pin 3 and Pin 6: not used

7.4 WiFi Communication Port

Two types of WiFi port are available to choose from: a USB Type-A port or an M14 aviation waterproof connector port.

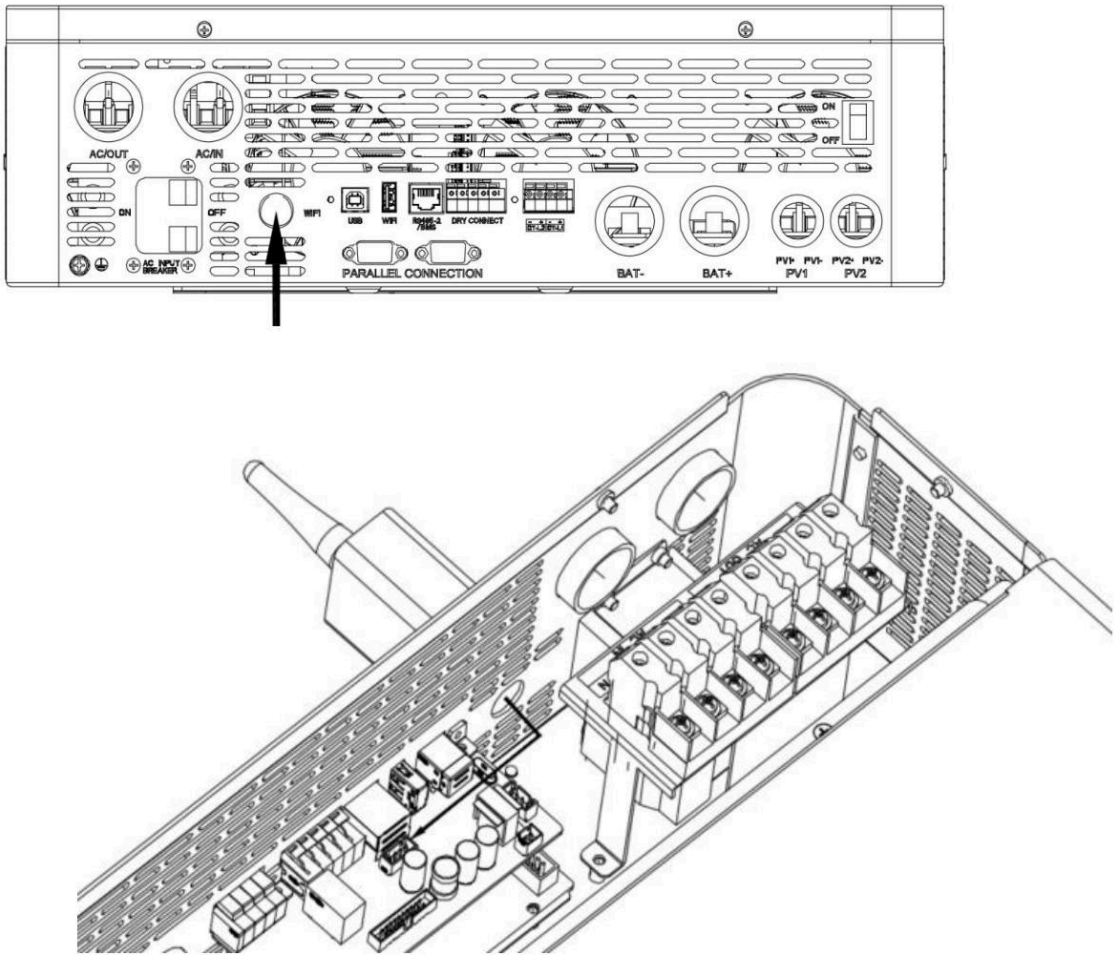
WiFi wiring and installation guidelines (WiFi optional):

Option 1 – USB Type-A WiFi module



WiFi module plugged into the USB Type-A WiFi port.

Option 2 – M14 aviation connector with antenna



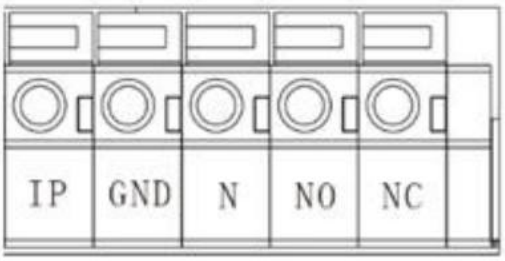
WiFi antenna on the M14 aviation connector port (arrow), with the connection inside the case.

7.5 Dry Contact Port

N-NO-NC function

How it works: This dry contact can turn a diesel generator ON and OFF.

1. Normally, the NC-N contact is closed (connected) and the NO-N contact is open (disconnected).
2. When the battery voltage reaches the low voltage disconnection point, the relay coil is energized. The NO-N contact closes and the NC-N contact opens. At this point, the NO-N contact can drive resistive loads rated at: 125VAC/1A, 230VAC/1A or 30VDC/1A.



Dry contact terminals: IP, GND, N, NO, NC

IP-GND function

How it works: This dry contact can turn the inverter's AC output power ON and OFF by remote control.

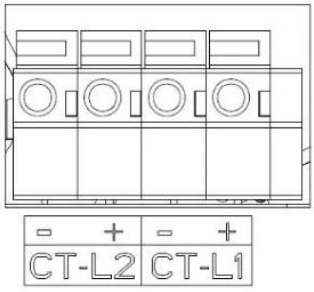
1. IP-GND open (disconnected) → AC output ON.
2. IP-GND closed (connected) → AC output OFF.

7.6 CT Sampling Detection (Anti Feed-in)

CT-L1 / CT-L2 function

This is an optional feature.

How it works: The unit measures (samples) the input line current to run the self-use function. Use only the CT sensors supplied for this inverter.



CT terminals: CT-L2 (− +) and CT-L1 (− +)

> **WARNING** This inverter does not have grid-connection certification. This function is only for off-grid use. Using it in a grid-connected system may violate local laws and utility rules. Check with a qualified professional in your area first.

7.7 Parallel Communication Function (Parallel Operation Only)

The KR-A12US has this feature:

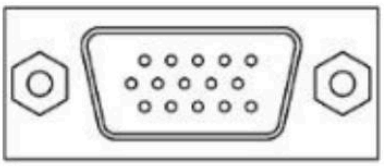
a. This port is for parallel communication. The paralleled units talk to each other through it.

b. Each inverter has two DB15 ports: one male connector and one female connector.

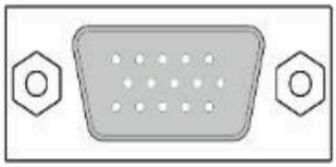
c. When connecting, connect the male connector of this inverter to the female connector of the inverter it is paralleled with. Or connect the female connector of this inverter to the male connector of the inverter it is paralleled with.

d. Do not connect an inverter's male connector to its own female connector.

Female connector



Male connector

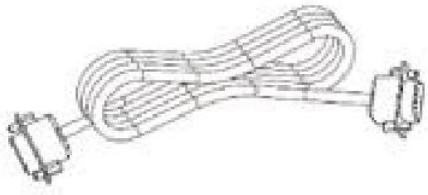


DB15 female connector (left) and male connector (right).

8. Parallel Installation and Wiring

8.1 Introduction

1. You can run up to six inverters in parallel. The KR-A12US supports parallel operation.
2. To use the parallel function, you must connect the part below firmly and securely. It comes in the package as an accessory.



Parallel communication line ×1 (included)

8.2 Precautions for Connecting the Parallel Lines

WARNING Follow every rule in items 1 to 6 below when you connect the parallel lines.

1. **Battery wiring**
   Parallel connection of batteries (set this on the screen in the "Parallel" menu):
   - Connect all inverters to the same battery.
   - Connect BAT+ to BAT+ and BAT− to BAT−.
   - Before you power on and start up, check that the wiring is correct. All cables must have the same length and the same wire diameter.
   - This prevents abnormal output from the parallel system caused by wrong wiring.

   Separate connection of batteries (set this on the screen in the "Parallel" menu):
   - Connect each inverter to its own battery.
   - Connect BAT+ to BAT+ and BAT− to BAT−.
   - Before you power on and start up, check that the wiring is correct. All cables must have the same length and the same wire diameter.
   - This prevents abnormal output from the parallel system caused by wrong wiring.
2. **AC OUT wiring**
   Parallel connection in split phase:
   - On all inverters, connect L1 to L1, L2 to L2, N to N, and PE to PE.
   - Before you power on and start up, check that the wiring is correct. All cables must have the same length and the same wire diameter.
   - This prevents abnormal output from the parallel system caused by wrong wiring.
   - For the exact wiring, see the wiring diagrams in Section 8.3.

Parallel connection in single phase:

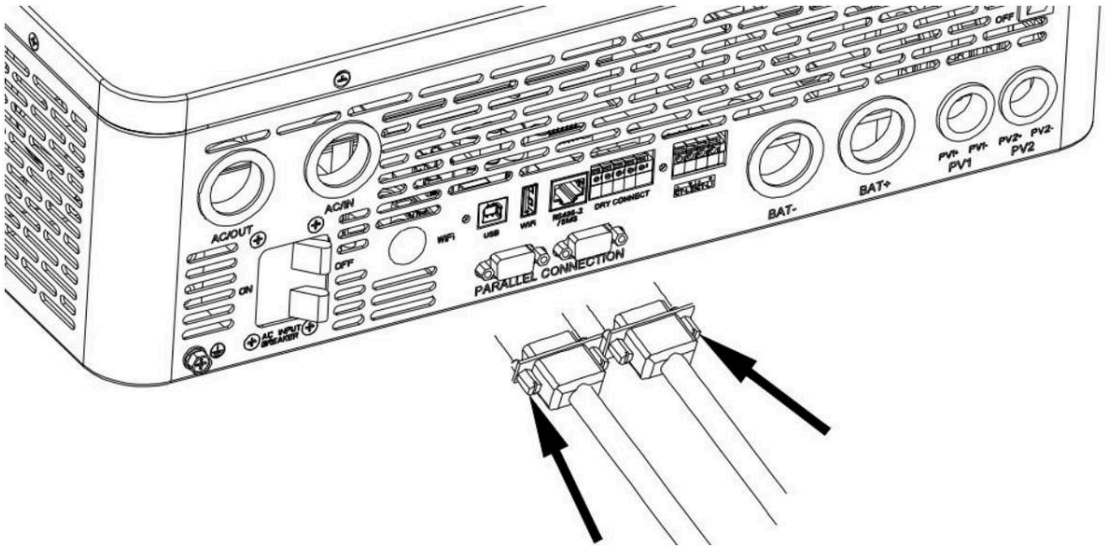
- On every inverter, connect N to N and PE to PE.
- On all inverters connected to the same phase, connect the L lines (L1 and L2) together.
- All other precautions are the same as for parallel connection in split phase.
- For the exact wiring, see the wiring diagrams in Section 8.3.

3. **AC IN wiring**
   Parallel connection in split phase:
   - On all inverters, connect L1 to L1, L2 to L2, N to N, and PE to PE.
   - Before you power on and start up, check that the wiring is correct. All cables must have the same length and the same wire diameter.
   - This prevents abnormal output from the parallel system caused by wrong wiring.
   - Do not connect different AC sources. This can damage equipment outside the inverter.
   - All inverters must share one AC source, and it must be the same source for every unit.
   - For the exact wiring, see the wiring diagrams in Section 8.3.

   Parallel connection in single phase:
   - On every inverter, connect N to N and PE to PE.
   - On all inverters connected to the same phase, connect the L lines (L1 and L2) together.
   - All other precautions are the same as for parallel connection in split phase.
   - For the exact wiring, see the wiring diagrams in Section 8.3.
4. **Wiring the parallel communication line**
   Parallel connection in single or split phase:
   - The parallel communication line is a standard DB15 computer cable. It is shielded.
   - Follow the "one-in-one-out" rule on each inverter. Connect the male connector (out) of one inverter to the female connector (in) of the next inverter in the parallel system.
   - Do not connect an inverter's male connector to its own female connector.
   - Tighten the communication line on each inverter with the DB15 connector's built-in end screws.
   - A loose or poorly connected communication line can cause abnormal output or damage the system.
5. Before and after you connect the system, check the system wiring diagrams below carefully. Make sure all wiring is correct and secure before you power on.
6. If you need to add a new inverter after the system is wired, powered on and running normally, follow these steps:
   - Disconnect the battery input, PV input, AC input and AC output.
   - Make sure every inverter is powered off.
   - Only then connect the new inverter into the system.

8.3 Parallel Connection Wiring Diagrams

1. After you connect the parallel communication lines, lock each one in place with its screws. See the diagram below.

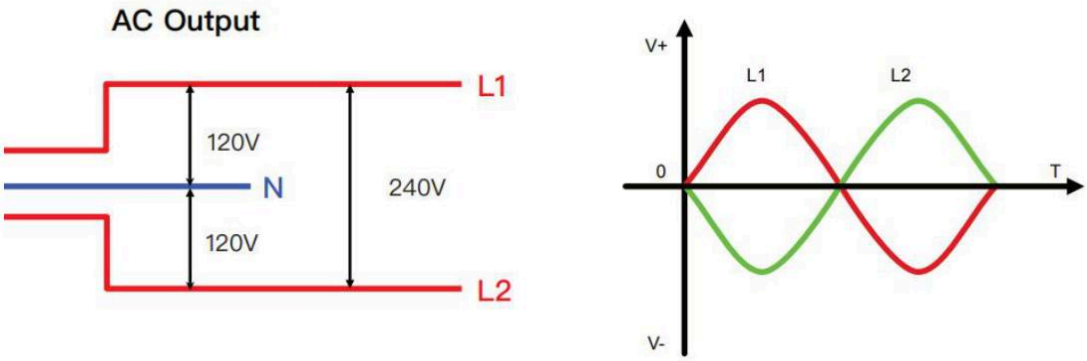


Lock the parallel communication lines to the PARALLEL CONNECTION ports with their screws (arrows).

2. The diagrams below show how to connect several inverters in parallel.

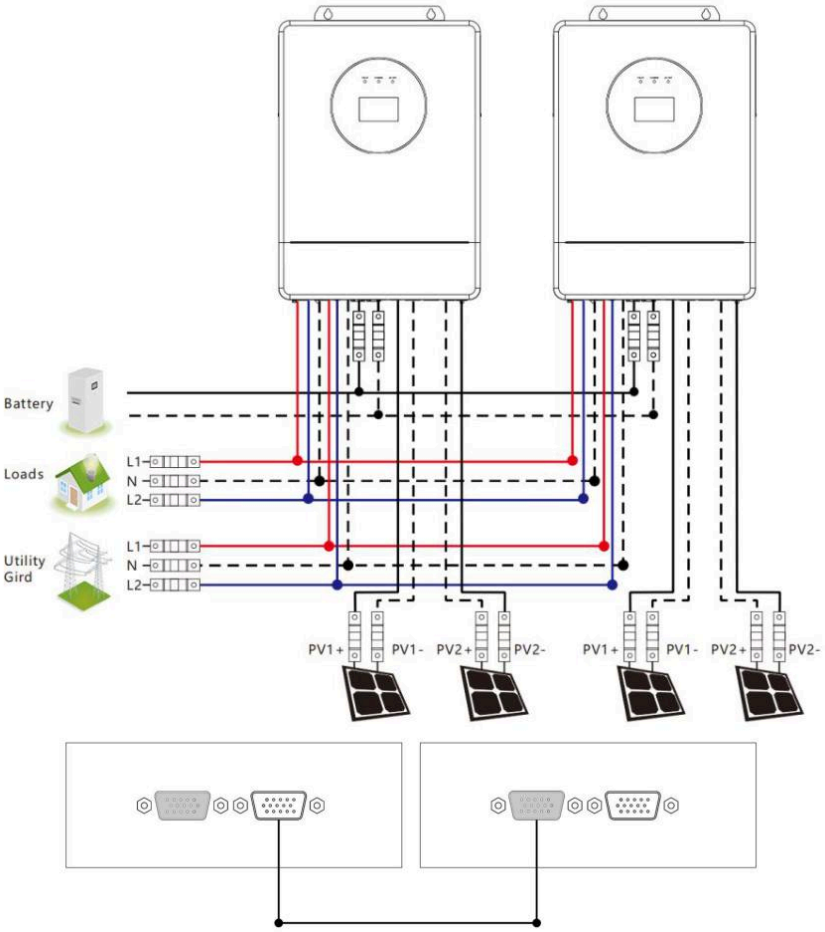
In the drawings below, "Utility Gird" is a misprint of "Utility Grid."

Parallel Connection in Split Phase



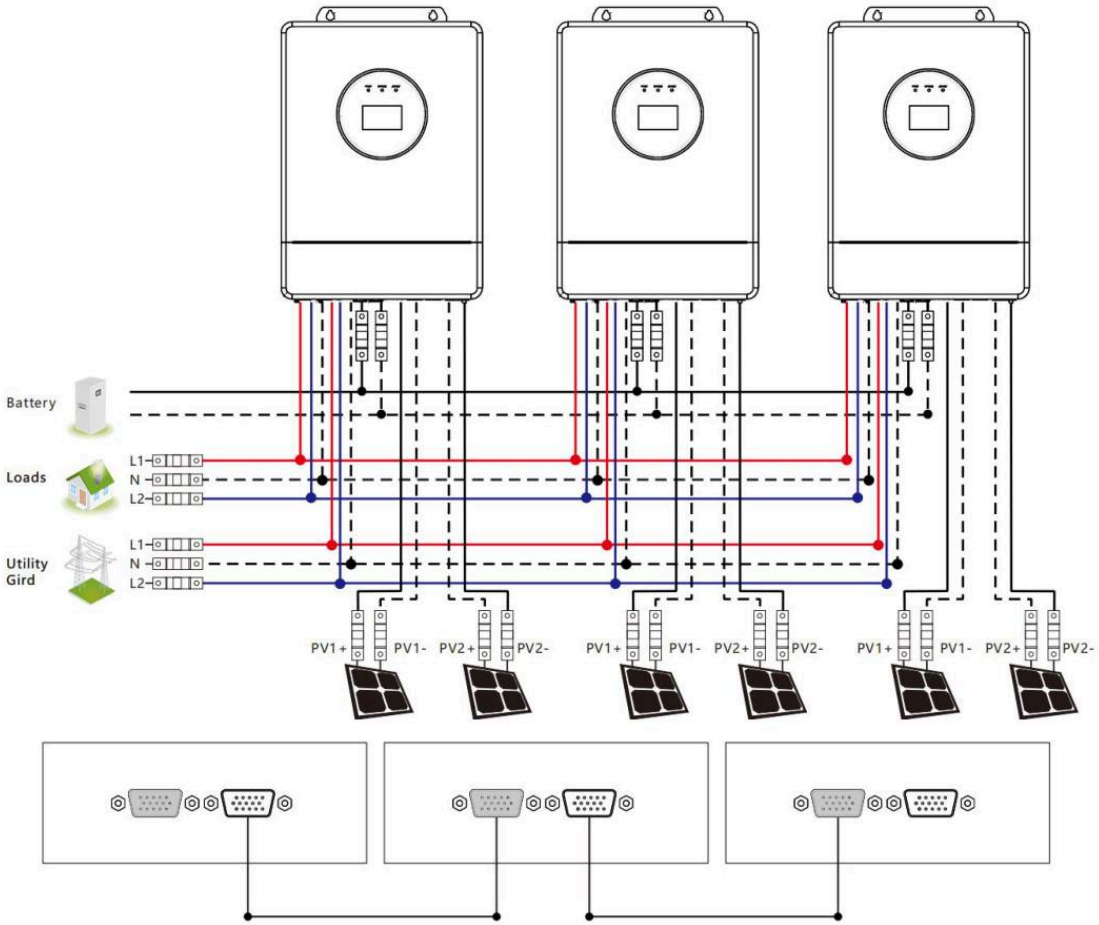
Split-phase AC output: L1–N 120V, L2–N 120V, L1–L2 240V

a) Two inverters connected in split-phase parallel



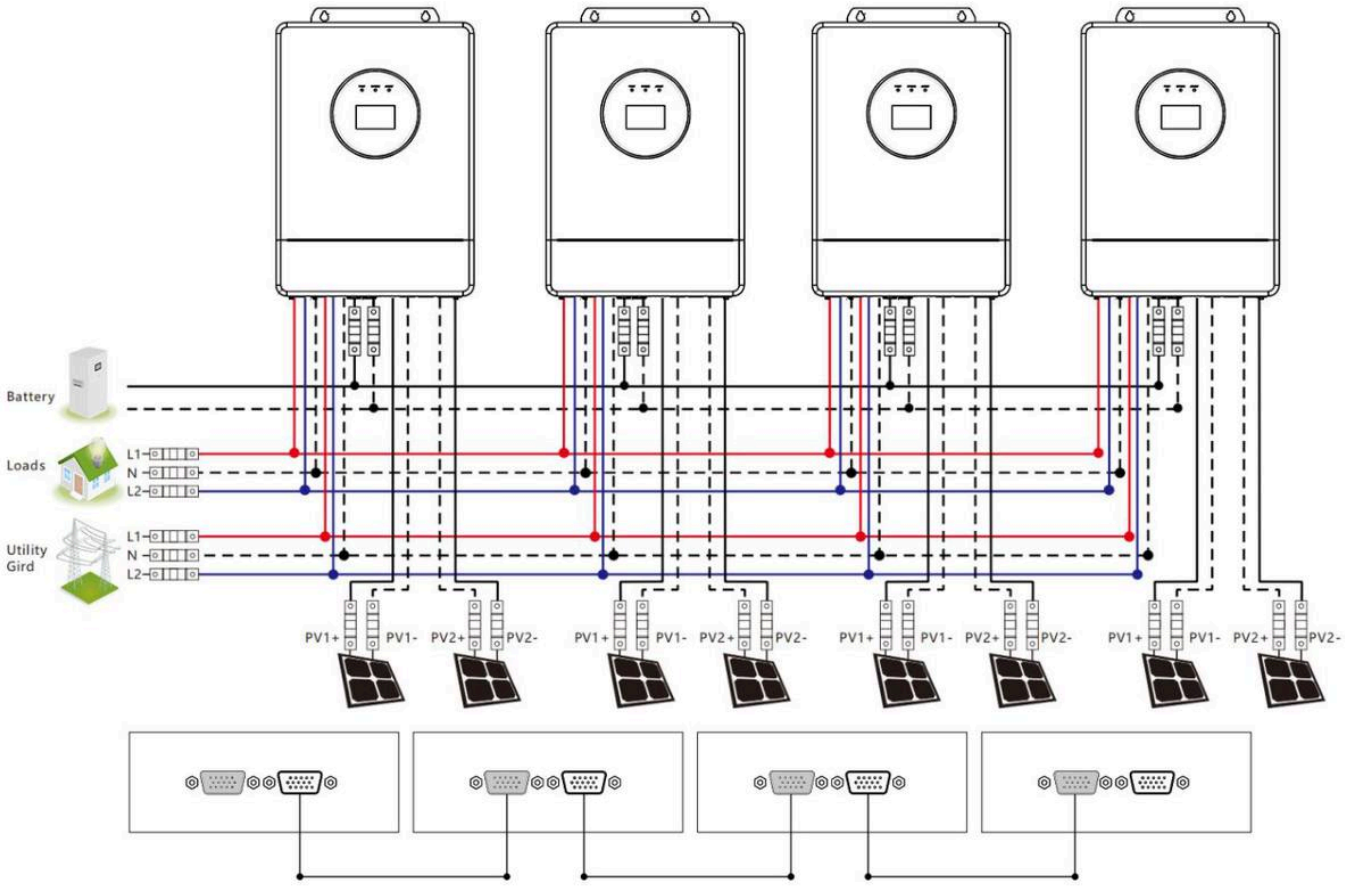
Two inverters, split-phase parallel

b) Three inverters connected in split-phase parallel



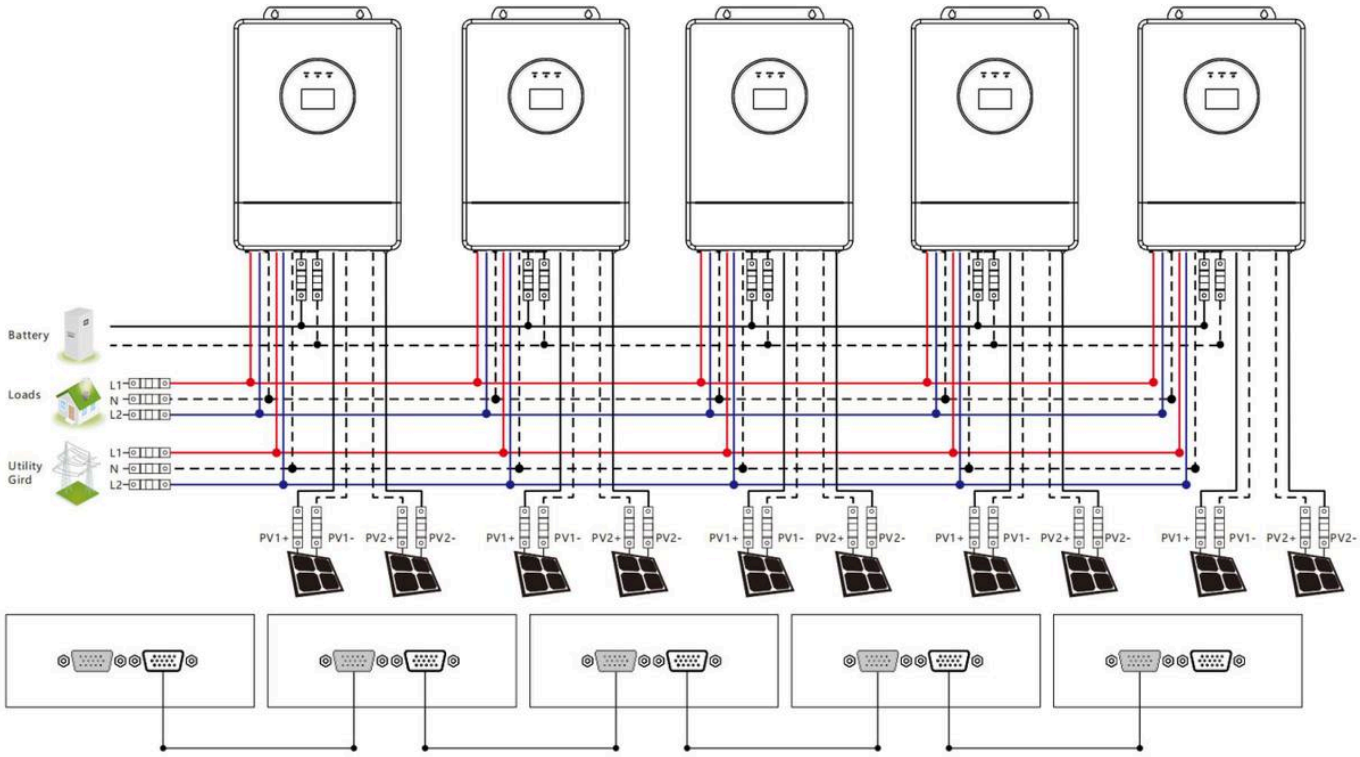
Three inverters, split-phase parallel

c) Four inverters connected in split-phase parallel



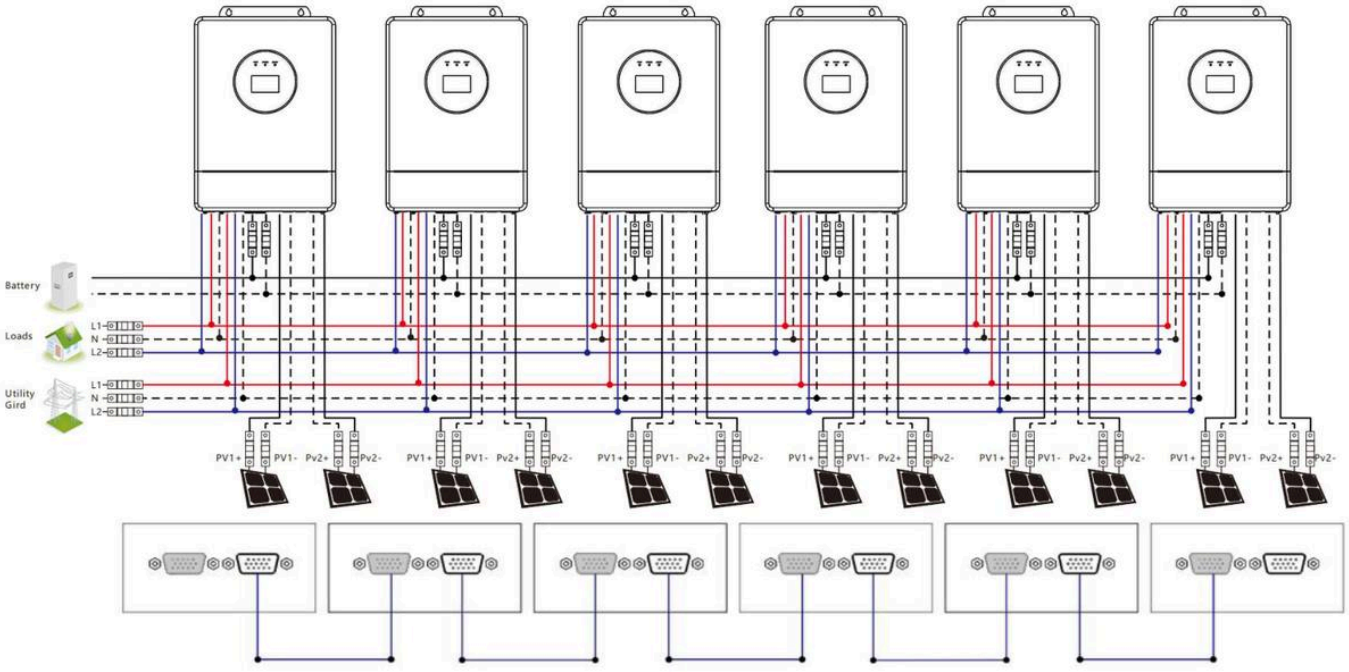
Four inverters, split-phase parallel

d) Five inverters connected in split-phase parallel



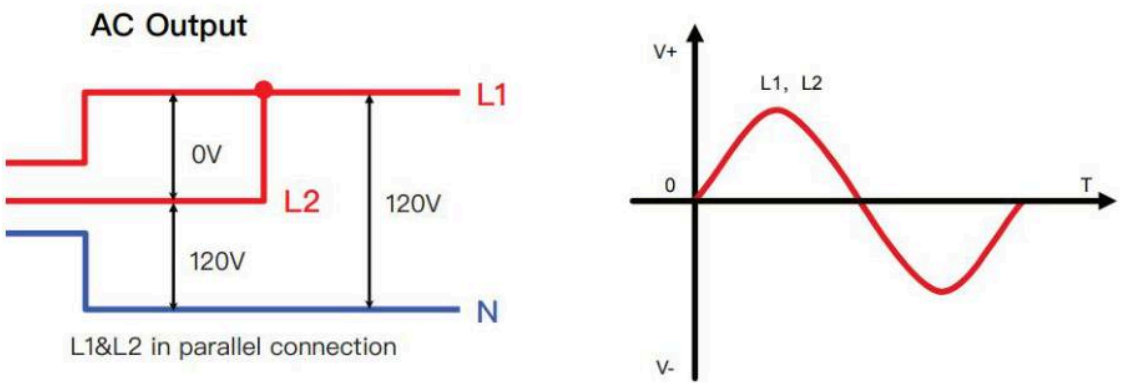
Five inverters, split-phase parallel

e) Six inverters connected in split-phase parallel



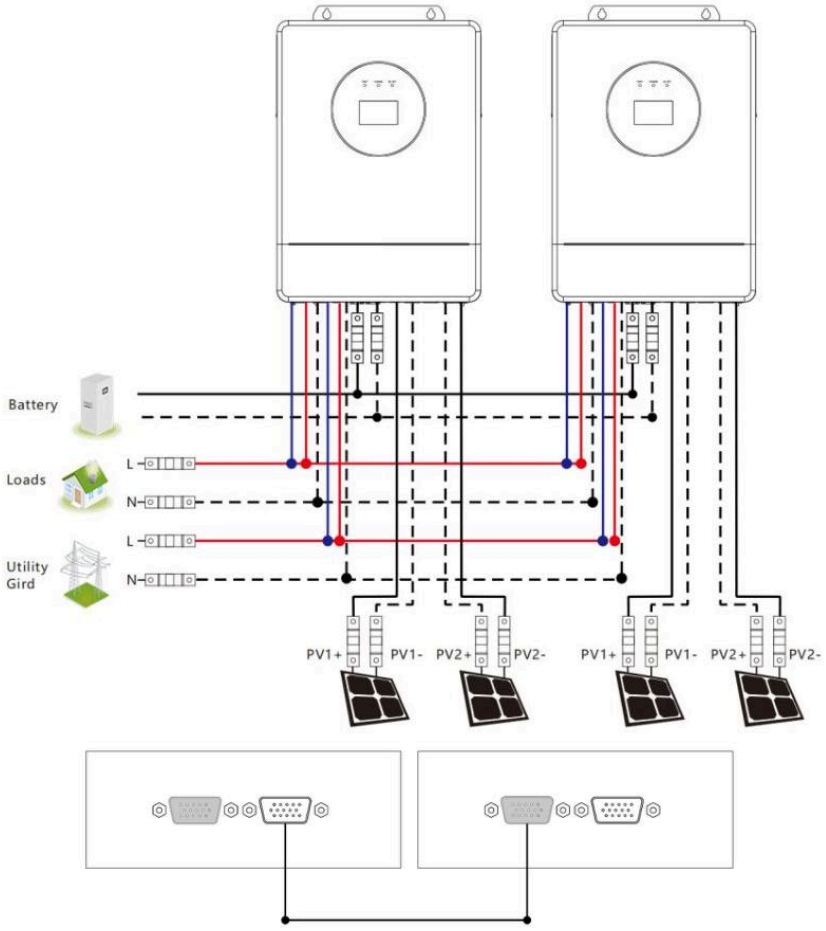
Six inverters, split-phase parallel

Parallel Connection in Single Phase



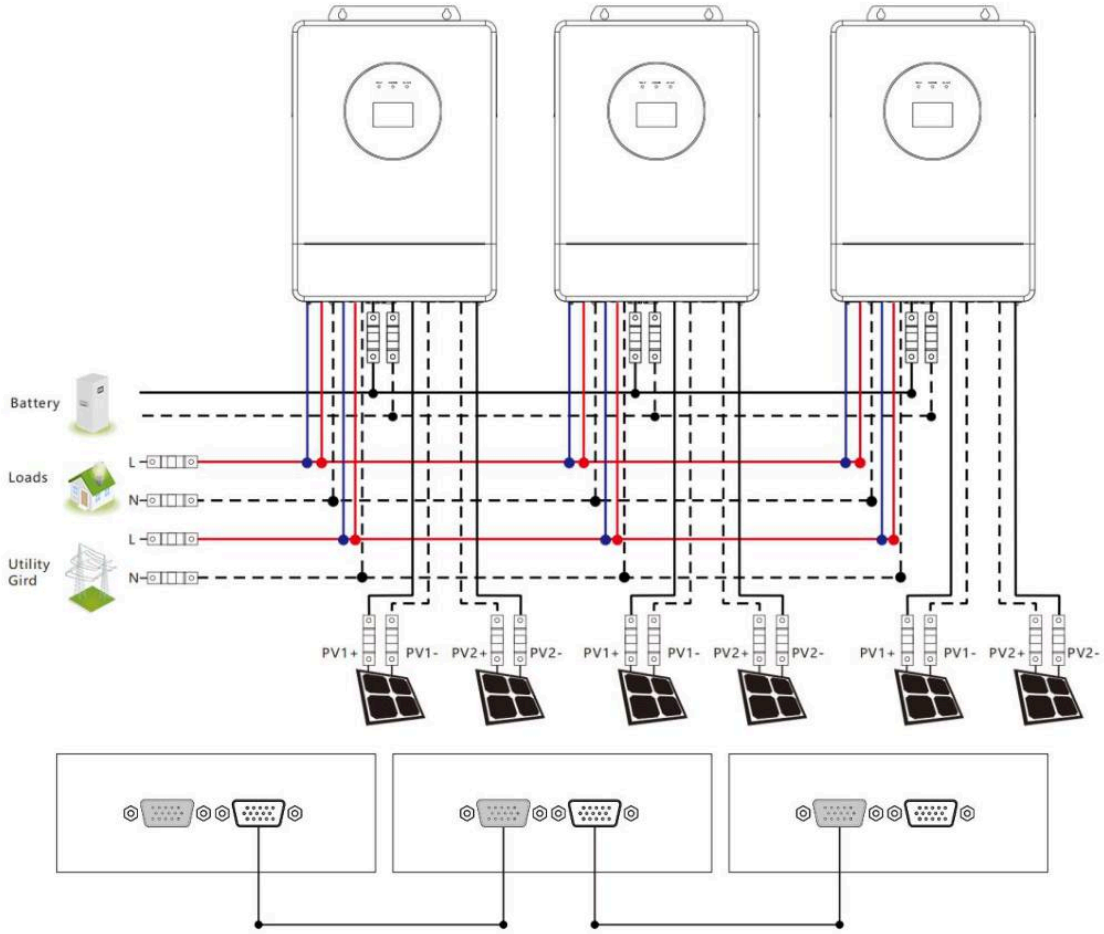
Single-phase AC output: L1 and L2 in parallel connection. L1–L2 0V, L1/L2–N 120V.

a) Two inverters connected in single-phase parallel



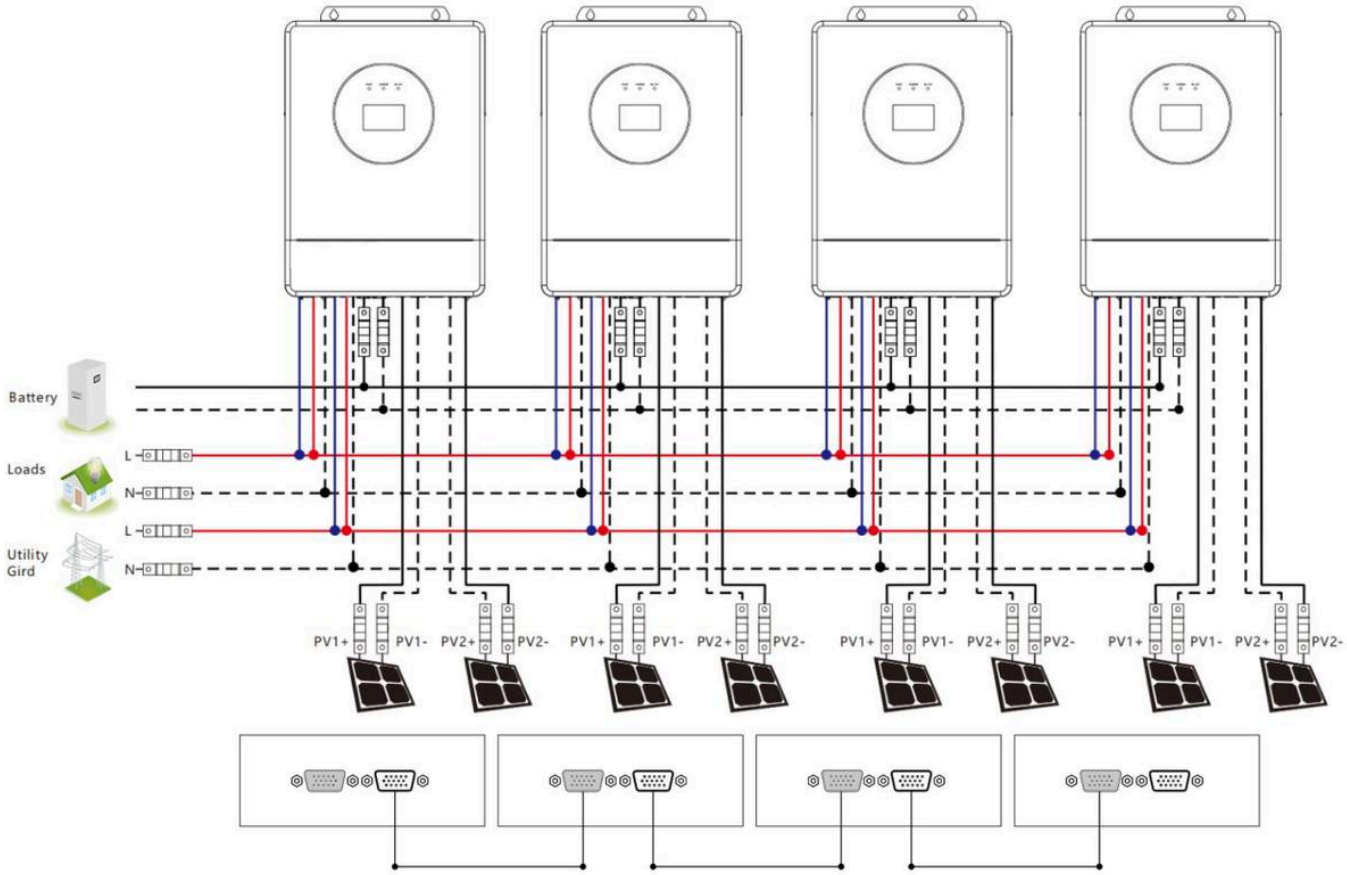
Two inverters, single-phase parallel

b) Three inverters connected in single-phase parallel



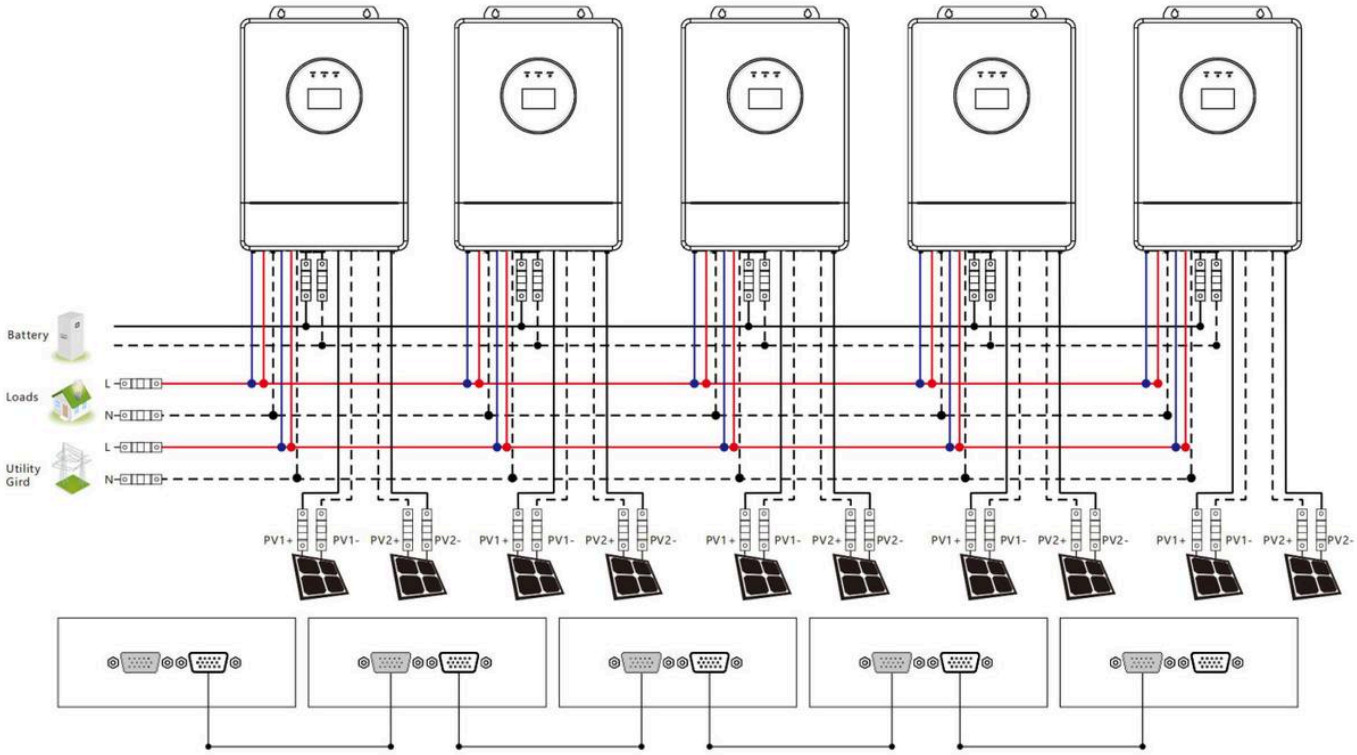
Three inverters, single-phase parallel

c) Four inverters connected in single-phase parallel



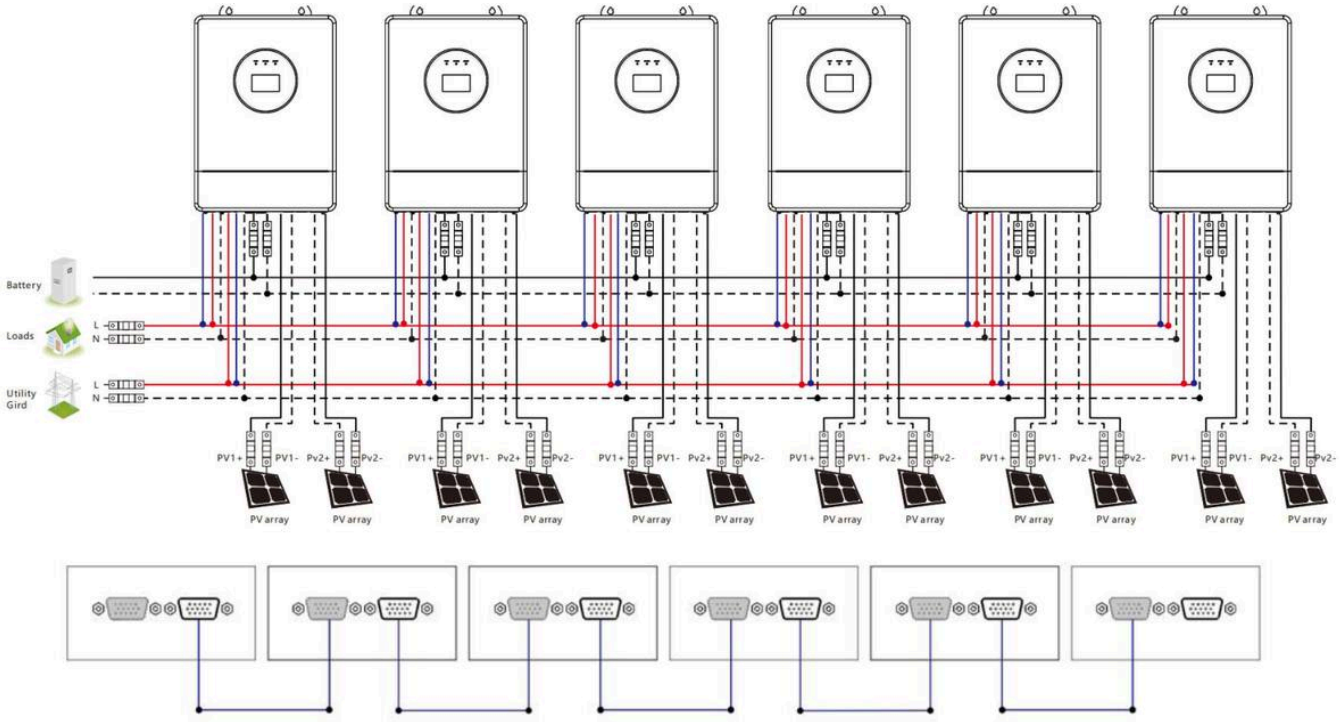
Four inverters, single-phase parallel

d) Five inverters connected in single-phase parallel



Five inverters, single-phase parallel

e) Six inverters connected in single-phase parallel



Six inverters, single-phase parallel

Notes

1. Before you start the system, check that every connection is correct. This helps prevent system problems.
2. All wiring must be fastened securely so no wire comes loose during use.
3. When you wire the AC output to the loads, follow the requirements of the load equipment. This prevents damage to the load equipment.
4. Set the AC output voltage to the same value on every unit, or set it on the host unit only. While the system runs, the host's voltage setting is used. The host (master) overwrites the setting on the other (slave) units so they all match. You can change this setting only in standby mode.

Setting	Default	Description
AC output voltage setting	120Vac **(default)**	Can be set to 100Vac, 105Vac, 110Vac, 120Vac or 127Vac. Default: 120V. The rated output power is reduced to: (Power Rate) × (Vset / 120).

5. When you use the parallel or split-phase function, set the items below to match your system.

Setting	Option	Description
AC output phase setting	Same phase	Single-phase mode. L1 and L2 are in the same phase (0 degrees). L1/L2 are in parallel connection. With the AC output voltage set to 120Vac, L1-N/L2-N voltage is 120Vac and L1-L2 voltage is 0Vac. You can change this only when the output is switched off.
	Split 120 P-seq	Split-phase mode. The L1-L2 phase difference is positive 120 degrees. L1/L2 are in split connection. With the AC output voltage set to 120Vac, L1-N/L2-N voltage is 120Vac and L1-L2 voltage is 208Vac. You can change this only when the output is switched off.
	Split 180	Split-phase mode. The L1-L2 phase difference is 180 degrees. L1/L2 are in split connection. With the AC output voltage set to 120Vac, L1-N/L2-N voltage is 120Vac and L1-L2 voltage is 240Vac. You can change this only when the output is switched off.
	Split 120 N-seq	Split-phase mode. The L1-L2 phase difference is negative 120 degrees. L1/L2 are in split connection. With the AC output voltage set to 120Vac, L1-N/L2-N voltage is 120Vac and L1-L2 voltage is 208Vac. You can change this only when the output is switched off.

6. To set the phase sequence on the screen, set one unit at a time. Turn on only the unit you are setting, and keep the other units turned off. After all units are set, power the system off and start it up again.
7. After the system starts, measure the output voltage and confirm it is correct. Then connect the loads.

9. Failure Codes and Countermeasures

9.1 Fault Codes

This table lists every fault code the unit can show. The "Affects output?" column tells you whether the fault turns off or limits the output.

Fault code	Fault name	Affects output?	Description
[01]	**Battery under voltage warning**	No	The battery voltage is lower than the "battery under voltage alarm" setting. The battery is in an under-voltage state.
[02]	**Battery under voltage protection**	Yes	The battery voltage is low. The unit turns off the output to stop the battery from discharging.
[03]	**Average battery discharge current over current protection**	Yes	The average battery discharge current has been above the maximum battery input current for 1 minute. The unit turns off the output to stop the battery from discharging.
[04]	**Instantaneous battery discharge over current protection**	Yes	The instantaneous battery discharge current is higher than the unit's maximum instantaneous value. The unit turns off the output to stop the battery from discharging.
[05]	**Battery not connected**	Yes	Warning: the battery is not connected.
[06]	**Battery over voltage**	Yes	The battery voltage is above the limit for the selected battery type or the set battery voltage. The unit turns off the output to stop battery charging.
[07]	**BMS low battery alarm**	No	The BMS sends a low battery alarm. (BMS enable must be set for this to work.)
[08]	**BMS low battery protection**	Yes	The BMS reports that the battery capacity is low. The unit turns off the output to stop the battery from discharging. (BMS enable must be set for this to work.)
[09]	**Bypass overload protection**	Yes	The grid/utility (mains) side is overloaded. The unit turns off the AC output and stops charging from the grid.
[10]	**Battery output overload protection**	Yes	The inverter is overloaded while running from the battery. The unit turns off the AC output and stops battery discharge.
[11]	**Battery inverter output short circuit**	Yes	The AC output of the battery inverter is short-circuited while discharging. The unit turns off the AC output and stops battery discharge.
[12]	**The AC output of the battery inverter over circuit**	Yes	The AC output of the battery inverter is "over circuit" while discharging. The unit turns off the AC output and stops battery discharge.
[13]	**The DC component of the battery inverter voltage is abnormal**	Yes	The DC component of the battery inverter voltage is abnormal. The unit turns off the AC output and stops battery discharge.
[14]	**Bus over voltage hardware sampling protection**	Yes	The bus voltage of the internal battery boost circuit is too high (hardware protection). The unit turns off the AC output and charging.

Fault code	Fault name	Affects output?	Description
[15]	**Bus over voltage software sampling protection**	Yes	The bus voltage of the internal battery boost circuit is too high (software protection). The unit turns off the AC output and charging.
[16]	**Bus under voltage protection**	Yes	The bus voltage of the internal battery boost circuit is too low (under-voltage protection). The unit turns off the AC output and charging.
[17]	**Bus short circuit protection**	Yes	The bus of the internal battery boost circuit has a short circuit (short-circuit protection). The unit turns off the AC output and charging.
[18]	**The PV input voltage is over voltage**	Yes	Protection: the solar input voltage is above the maximum allowed input voltage.
[19]	**PV Over Curr Sw**	Yes	Solar charging over-current software protection. Solar charging turns off.
[20]	**PV over Curr Hw**	No	Solar charging over-current hardware protection. Solar charging turns off.
[21]	**PV Iso Low**	Yes	The PV insulation impedance is low. The inverter shuts down.
[22]	**The PV heat sink is overheated**	No	The solar charging heat sink (radiator) is too hot. Solar charging turns off.
[23]	**The AC heat sink is overheated**	Yes	The heat sink is too hot. The unit turns off AC charging or battery inverter discharge.
[24]	**The temperature of the main transformer is overheated**	Yes	The internal main transformer is too hot. The unit turns off AC charging or battery inverter discharge.
[25]	**AC input relay short circuit**	Yes	AC input relay short-circuit protection. It keeps the inverter's AC output from being pushed back into the bypass AC input. The inverter turns off.
[26]	**Output relay short**	Yes	Output relay short circuit.
[27]	**Fan failure**	Yes	The fan is blocked or has failed. The unit turns off the inverter output and charging.
[28]	**EEPROM**	Yes	The program is not activated.
[29]	**SPI Comm Err**	Yes	Communication error between the master chip and the slave chip.
[30]	**Type detection error**	Yes	The model was not set before delivery, and the model identification is wrong.
[31]	**Bus soft start fail**	Yes	Bus soft start failure.
[32]	**Leakage Curr Over**	Yes	The leakage current sensor has a sampling error.
[33]	**Parallel control CAN faulty**	Yes	In parallel mode, CAN control is lost. AC output and charging turn off.
[34]	**Parallel communication CAN fault**	Yes	In parallel mode, CAN communication is lost. AC output and charging turn off.
[35]	**Parallel mode is faulty**	Yes	In parallel mode, the units in the system have different parallel mode settings.
[36]	**Parallel current sharing fault**	Yes	In parallel mode, the battery inverter AC output current is very uneven from unit to unit. AC output and charging turn off.

Fault code	Fault name	Affects output?	Description
[37]	**Parallel ID setting error**	Yes	In parallel mode, the RS485 addresses conflict again and again. This fault stops AC output and charging. After the host automatically reassigns the addresses, the fault clears and the host enters parallel operation.
[38]	**Inconsistent battery in parallel mode**	Yes	In parallel mode, the battery input voltage is very different from unit to unit.
[39]	**Inconsistent AC input source in parallel mode**	Yes	In parallel mode, the units do not share the same AC input source.
[40]	**The parallel mode synchronization fails**	Yes	In parallel mode, the hardware synchronization signal was not received. Parallel operation and AC output stop.
[41]	**Inconsistent system firmware version in parallel mode**	Yes	The program versions in the parallel system do not match. Parallel operation and AC output stop.
[42]	**The parallel communication cable is faulty**	Yes	The parallel communication line is faulty. Parallel operation and AC output stop.
[43]	**Serial number error**	Yes	The serial number was not set at the factory, or the same factory serial number is set more than once in the system.
[44]	**Batt cap not enough**	Yes	The battery voltage of the connected unit is low.
[45]	**Aux DSP abnormal**	Yes	Contact KOOR support.
[46]	**Bus unbalance**	Yes	There is an imbalance between the positive and negative busbars.
[47]	**External CT Host Error**	No	In a parallel system, only one unit on each phase can be set as the external CT host.
[48]	**Mains phase err**	No	Check the grid/utility (mains) phase.
[49]	**BMS communication error**	No	Check that the communication line is connected correctly. Check that the **BMS** is set to the matching lithium battery communication protocol.
[50]	**BMS other alarm**	No	Check the BMS fault type and troubleshoot the lithium battery problem.
[51]	**BMS battery over temperature alarm**	No	BMS alarm: lithium battery over temperature.
[52]	**BMS battery over current alarm**	No	BMS alarm: lithium battery over current.
[53]	**BMS battery over voltage alarm**	No	BMS alarm: lithium battery over voltage.
[54]	**BMS battery low voltage alarm**	No	BMS alarm: lithium battery low voltage.
[55]	**BMS battery low temperature alarm**	No	BMS alarm: lithium battery low temperature.

9.2 Troubleshooting

Fault code	Fault	What to do
/	**No display on the screen**	Check that the battery circuit breaker or the PV circuit breaker is closed. Check that the switch is in the "ON" position. Tap the screen to wake it.
[06]	**Battery over voltage**	Measure the battery voltage to see if it is above the rated value. If it is, discharge the battery until its voltage is below the battery over-voltage recovery point.
[01] [02]	**Battery under voltage protection**	Charge the battery until it reaches the low voltage disconnection recovery voltage.
[27]	**Fan failure**	Check whether the fan is not turning or is blocked by a foreign object.
[22] [23]	**Heat sink over temperature protection**	When the unit cools below the recovery temperature, normal charge and discharge control starts again.
[09] [10]	**Bypass overload protection, inverter overload protection**	1. Use fewer power devices (reduce the load). 2. Restart the unit to restore load output.
[11]	**Inverter short circuit protection**	1. Check the load wiring carefully and clear any short-circuit points. 2. Power the unit up again to restore load output.
[18]	**PV over-voltage**	Use a multimeter to check whether the PV input voltage is above the rated maximum allowed input voltage.
[05]	**Battery not connected**	Check whether the battery is not connected, or whether the battery circuit breaker is not closed.
[40] [42]	**Parallel connection fault**	Check whether the parallel cable is poorly connected, for example loose or wired wrong.
[37]	**Parallel ID setting error**	Check whether the same parallel ID number is set more than once.
[36]	**Parallel current sharing fault**	Check whether the parallel current sharing cable is poorly connected, for example loose or wired wrong.
[39]	**Inconsistent AC input source in parallel mode**	Check whether the parallel AC inputs come from the same input interface. In three-phase mode, if two of the three phases report that the grid input sources do not match, swap the grid input wiring of any two units. This checks whether the grid input phase sequence is different from the set phase sequence.
[41]	**Inconsistent system firmware version in parallel mode**	Check whether every inverter has the same software version.

NOTE If a fault cannot be fixed with the steps in the table above, contact KOOR support at 574-218-1973 for technical help. Do not take the equipment apart yourself.

10. Protection and Maintenance

10.1 Built-in Protections

No.	Protection	Description
1	**PV current/power limiting protection**	If the charging current or power of the connected PV array is above the PV rating, the unit charges at the rated value.
2	**PV night reverse-current protection**	At night the battery voltage is higher than the PV module voltage. This protection keeps the battery from discharging through the PV module.
3	**Mains input over voltage protection**	When the grid/utility (mains) voltage is above 140V, grid charging stops and the unit switches to inverter mode.
4	**Mains input under voltage protection**	When the grid/utility (mains) voltage is lower than or equal to 85V, grid charging stops and the unit switches to inverter mode.
5	**Battery over voltage protection**	When the battery voltage reaches the over voltage disconnection point, PV and grid charging of the battery stop automatically. This keeps the battery from being overcharged and damaged.
6	**Battery low voltage protection**	When the battery voltage reaches the low voltage disconnection point, battery discharging stops automatically. This keeps the battery from being over-discharged and damaged.
7	**Load output short circuit protection**	If a short circuit at the load output terminal lasts more than 200 milliseconds, the AC output turns off at once.
8	**Heat sink over temperature protection**	When the internal temperature is too high, the inverter stops charging and discharging. When the temperature returns to normal, charging and discharging start again.
9	**Overload protection**	After an overload protection, the output turns back on after 3 minutes. After 5 overload protections in a row, the output stays off until the unit is powered up again. For the exact overload levels and times, see the technical parameters table (Chapter 11).
10	**PV reverse polarity protection**	If the PV polarity is reversed, the unit will not be damaged.
11	**AC reverse protection**	Keeps battery inverter AC current from flowing backward into the Bypass.
12	**Bypass over current protection**	Built-in AC input over-current protection circuit breaker.
13	**Battery input over current protection**	When the battery discharge output current stays above the maximum value for 1 minute, the AC input is switched to the load.
14	**Battery input protection**	If the battery is connected backward or the inverter is short-circuited, the battery input fuse inside the inverter blows. This keeps the battery from being damaged or causing a fire.
15	**Charge short protection**	If the external battery port is short-circuited while charging from PV or AC, the inverter protects itself and stops the output current.
16	**CAN communication loss protection**	In parallel operation, the unit gives an alarm when CAN communication is lost.
17	**Parallel connection error protection**	In parallel operation, the equipment is protected when the parallel line is lost.
18	**Parallel battery voltage difference protection**	In parallel operation, the equipment is protected when the battery connections do not match and the battery voltage is very different from what the host detects.

No.	Protection	Description
19	**Parallel AC voltage difference protection**	In parallel operation, the equipment is protected when the AC IN input connections do not match.
20	**Parallel current sharing fault protection**	In parallel operation, the running equipment is protected when the load on each inverter is very different. This can be caused by a poor current sharing line connection or a damaged device.
21	**Synchronization signal fault protection**	The equipment is protected when there is a fault in the guidance signal between the parallel buses, which makes the inverters behave differently.

10.2 Maintenance

1. For the best long-term performance, do the following checks twice a year.
2. Make sure nothing blocks the airflow around the unit. Remove any dirt or debris from the heat sink.
3. Check that no exposed wires are damaged by sunlight, rubbing against nearby objects, drying out, insect or rodent bites, and so on. Repair or replace the wires if needed.
4. Check that the indicators and display match how the unit is working. Watch for any faults or errors on the display, and fix them if needed.
5. Check all wiring terminals for corrosion, damaged insulation, signs of high heat, or burning/discoloration. Tighten the screws.
6. Check for dirt, insect nests and corrosion. Clean as needed.
7. If the arrester has failed, replace it right away. This prevents lightning damage to the unit and even to the user's other equipment.

NOTICE **KOOR does not accept any liability for damage caused by:**

a. Improper use, or use in an unsuitable location.

b. A PV module open-circuit voltage that is above the rated maximum allowed voltage.

c. An operating temperature above the allowed operating temperature range.

d. Taking apart or repairing the inverter without permission.

e. Force majeure, or damage during shipping or handling.

DANGER **Electric shock hazard! Before doing any of the work above, make sure all power sources to the inverter are disconnected and all capacitors are discharged. Only then check or work on the unit.**

11. Technical Parameters

Model	KR-A12US
Parallel mode	
Permitted parallel number	1~6
AC mode	
Rated input voltage	120 (L1/N, L2/N) / 240Vac (L1/L2)
Input phase voltage range	(85Vac~140Vac) ±2%
Frequency	50Hz/60Hz (auto detection)
Frequency range	47±0.3Hz~55±0.3Hz (50Hz); 57±0.3Hz~65±0.3Hz (60Hz)
Overload/short circuit protection	Circuit breaker
Efficiency	>95%
Conversion time (bypass and inverter)	10ms (typical)
AC reverse protection	Yes
Maximum bypass phase overload current	2-pole, 60A/60A (L1/L2)
Inverter mode	
Output voltage waveform	Pure sine wave
Rated output power (VA)	Split phase: 12000
	Single phase: 7200 (limited by the wiring terminals)
Rated output power (W)	Split phase: 12000
	Single phase: 7200 (limited by the wiring terminals)
Power factor	1
Rated output voltage (Vac)	120Vac (L1/N, L2/N) / 240Vac (L1/L2)
Output voltage error	±5%
Output frequency range (Hz)	50Hz±0.3Hz / 60Hz±0.3Hz
Maximum efficiency	>91%
Overload protection	(105% < load < 110%) ±5%: reports an error and turns off the output after 5 minutes; (110% < load < 125%) ±5%: reports an error and turns off the output after 10 seconds; Load >125% ±5%: reports an error and turns off the output after 5 seconds
Peak power	Split phase: 18000VA
	Single phase: 18000VA

Model	KR-A12US
Motor load capability	6HP
Output short circuit protection	Yes; protection acts after a continuous short circuit of 1 second
Bypass circuit breaker phase specification	2-pole, 63A/63A (L1/L2)
Rated battery input voltage	51.2V (minimum starting voltage 44V)
Battery voltage range	40.0Vdc~58.6Vdc ±0.6Vdc (the under-voltage alarm, shutdown, over-voltage alarm, over-voltage recovery and other points can be set on the LCD screen)
Power saving mode self-consumption	When the load on each phase is zero or less than 25W, the unit enters power-saving mode
AC charging	
Battery type	Lead acid or lithium battery
Maximum charge current (can be set)	160A
Charge current error	±5Adc
Charge voltage range	40–58.6Vdc
Short circuit protection	Circuit breaker and blown fuse
Circuit breaker phase specifications	2-pole, 60A/60A (L1/L2)
PV charging	
Number of MPPT groups	2
Maximum PV open circuit voltage	500Vdc
PV operating voltage range	90Vdc~485Vdc
MPPT voltage range	90Vdc~450Vdc
Battery voltage range	40-58.6Vdc
Maximum PV input power	9000W + 9000W
Maximum PV input current	27A + 27A
PV charge current (can be set)	250A
Charging short circuit protection	Blown fuse
Wiring protection	Reverse polarity protection
Hybrid charging max charger current specifications (AC charger + PV charger)	
Max charger current (can be set)	0-250A

Model	KR-A12US
Certified specifications	
Certification	See product label
EMC certification level	See product label
Operating temperature range	-10°C to 55°C (14°F to 131°F) (derating above 45°C / 113°F)
Storage temperature range	-25°C ~ 60°C (-13°F ~ 140°F)
Humidity range	5% to 95% (conformal coating protection)
Noise	≤60dB
Heat dissipation	Forced air cooling, variable fan speed
Communication interface	USB / RS485 (WiFi/GPRS) / dry contact control
Size (L*W*D)	620mm*450mm*132mm (24.41 in × 17.72 in × 5.20 in)
Weight	23.8 kg (52.47 lb)

Appendix A. Setting Up a KOOR Battery

This appendix covers the two KOOR lithium batteries: the **KOOR Active Stack** (stackable, 51.2V 314Ah, 16 kWh) and the **KOOR 16K Pro** tower. Both talk to this inverter on the **Pylon** protocol over **CAN**.

> Turn the inverter OFF and switch the battery off before you plug in or unplug a communication cable. Set the address switches with the battery off.

A.1 The Rules That Apply to Both

- **Only one battery talks to the inverter.** That one is the master and its address switches are **all OFF (address 0)**. Its CAN jack goes to the inverter's RS485-2/BMS port.
- **The rest chain in a line on RS485-2** — out of B on one battery, into A on the next. Never run a cable from the last one back to the master.
- **The cable to the inverter is a plain straight ethernet cable** for both batteries.
- **No two batteries share an address.**
- The inverter's BMS port has **CAN-H on pin 4 and CAN-L on pin 5**. RS485 sits on pins 1 and 2, repeated on pins 7 and 8.

A.2 Address Switches (both batteries)

Address	1	2	3	4
0 — master	off	off	off	off
1	**ON**	off	off	off
2	off	**ON**	off	off
3	**ON**	**ON**	off	off
4	off	off	**ON**	off
5	**ON**	off	**ON**	off
6	off	**ON**	**ON**	off
7	**ON**	**ON**	**ON**	off
8	off	off	off	**ON**
9	**ON**	off	off	**ON**
10	off	**ON**	off	**ON**
11	**ON**	**ON**	off	**ON**
12	off	off	**ON**	**ON**
13	**ON**	off	**ON**	**ON**
14	off	**ON**	**ON**	**ON**

Up is ON. Set the switches with the battery switched off. Up to 15 batteries: the master is 0, the rest 1 to 14. Changed a switch with the battery on? Hold **RST 3 s** (sleep), then **3 s again** (wake) so it reads the new address.

A.3 KOOR Active Stack

Go by the words stamped on the battery panel. The pair marked **RS485-1 / CAN** is the inverter pair: **CAN** (the right-hand jack) goes to the inverter, RS485-1 is the backup path. The dual jack marked **RS485-2** (A and B) is for chaining pack to pack. RS232 (RJ11) is for the battery software only.

Jack	What it is	Where it goes
CAN	CAN	Master only → inverter BMS port
RS485-1	RS485	Backup path to the inverter (see A.5)
RS485-2 — A and B	RS485	Pack to pack: out of B, into A on the next pack
RS232 (RJ11)	Service	Battery software only

Cable: a plain straight ethernet cable (CAT5e or CAT6), both ends wired the same. Pin 4 is CAN-H and pin 5 is CAN-L, which lines up with the inverter. The master's RS485-2 A jack and the last pack's B jack stay empty. RS485 runs at 9600 baud.

A.4 KOOR 16K Pro

The 16K Pro tower wires **exactly like the Active Stack**. Its panel reads ON/OFF, RST, ADD (the address switches), DRY, then the jacks **RS485-1**, **CAN**, **RS232** and **RS485-2** (a dual jack, A and B).

Jack	Where it goes
CAN	Master tower only → inverter BMS port. Every other tower's CAN jack stays empty.
RS485-2 — A and B	Tower to tower: out of B, into A on the next tower. Master A and last tower B stay empty.
RS485-1	Backup path to the inverter only. Empty on a normal job.
RS232 and DRY	Not used. A chain cable never goes in these.

> **Same cable as the Active Stack — a plain straight ethernet cable.** Nothing crossed, nothing custom, nothing to crimp.

A.5 Inverter Settings

Set these in order, then save.

Setting	Set it to	Why
Battery type	LFP16	16 cells in series, 51.2V nominal.
RS485-2	CAN-BMS	Turns on BMS communication over CAN.
BMS protocol	Pylon Can	Both KOOR batteries speak Pylon.

> **Backup path:** if there is still no battery data, you can try RS485 instead. Move the cable to the battery's **RS485-1** jack (a plain straight cable works there too), then set **RS485-2** to **485-BMS** and **BMS protocol** to **Pylon**.

> **Set Pylon on the battery too — not just the inverter.** Both ends must be on it, and some packs come set to a different protocol. On every pack set **CAN protocol = Pylon** (Active Stack: battery Bluetooth app, Settings, password **1234**), then hold RST 3 s, then 3 s again.

A.6 How to Tell It Is Talking

- The screen shows the battery **state of charge as a percent** from the BMS, with battery voltage and current.
- No BMS fault code is showing.
- Charge current follows the battery's own limit instead of the value you set by hand.

A.7 If It Will Not Talk

What you see	What to check, in this order
No battery data at all	Is the battery on Pylon? Set CAN protocol = Pylon on every pack and restart it. Then the cable — a plain straight ethernet cable in the CAN jack, not RS485-1 or RS232; swap in a known-good one. Then the master's address (all four switches OFF). Then the inverter protocol setting.
BMS communication fault code	Reseat the RJ45 at both ends until it clicks. Battery switched on with its breaker closed. Only one battery cabled to the inverter.
Data shows, then drops out	Two batteries on the same address, or the chain loops back to the master.
Charging stops early	Normal — the BMS is limiting charge. Check battery temperature and the charge stop settings.

Still stuck? Call KOOR support at 574-218-1973 with the inverter model, the battery model, and the fault code on the screen.